

Symmetric Products $R^T R$ and RR^T and LDL^T

Choose any matrix R , probably rectangular. Multiply R^T times R . Then the product $R^T R$ is automatically a square symmetric matrix:

$$\text{The transpose of } R^T R \text{ is } R^T (R^T)^T \text{ which is } R^T R. \quad (7)$$

That is a quick proof of symmetry for $R^T R$. We could also look at the (i, j) entry of $R^T R$. It is the dot product of row i of R^T (column i of R) with column j of R . The (j, i) entry is the same dot product, column j with column i . So $R^T R$ is symmetric.

The matrix RR^T is also symmetric. (The shapes of R and R^T allow multiplication.) But RR^T is a different matrix from $R^T R$. In our experience, most scientific problems that start with a rectangular matrix R end up with $R^T R$ or RR^T or both. As in least squares.

Example 4 Multiply $R = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$ and $R^T = \begin{bmatrix} -1 & 0 \\ 1 & -1 \\ 0 & 1 \end{bmatrix}$ in both orders.

$$RR^T = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \text{ and } R^T R = \begin{bmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix} \text{ are both symmetric matrices.}$$

The product $R^T R$ is n by n . In the opposite order, RR^T is m by m . Both are symmetric, with positive diagonal (*why?*). But even if $m = n$, it is not very likely that $R^T R = RR^T$. Equality can happen, but it is abnormal.

Symmetric matrices in elimination $A^T = A$ makes elimination faster, because we can work with half the matrix (plus the diagonal). It is true that the upper triangular U is probably not symmetric. **The symmetry is in the triple product $A = LDU$.** Remember how the diagonal matrix D of pivots can be divided out, to leave 1's on the diagonal of both L and U :

$$\begin{aligned} \begin{bmatrix} 1 & 2 \\ 2 & 7 \end{bmatrix} &= \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} && LU \text{ misses the symmetry of } A \\ &= \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} && LDU \text{ captures the symmetry} \\ &&& \text{Now } U \text{ is the transpose of } L. \end{aligned}$$

When A is symmetric, the usual form $A = LDU$ becomes $A = LDL^T$. The final U (with 1's on the diagonal) is the transpose of L (also with 1's on the diagonal). The diagonal matrix D containing the pivots is symmetric by itself.

If $A = A^T$ is factored into LDU with no row exchanges, then U is exactly L^T .

The symmetric factorization of a symmetric matrix is $A = LDL^T$.

Notice that the transpose of LDL^T is automatically $(L^T)^T D^T L^T$ which is LDL^T again. The work of elimination is cut in half, from $n^3/3$ multiplications to $n^3/6$. The storage is also cut essentially in half. We only keep L and D , not U which is just L^T .

Permutation Matrices

The transpose plays a special role for a *permutation matrix*. This matrix P has a single “1” in every row and every column. Then P^T is also a permutation matrix—maybe the same or maybe different. Any product $P_1 P_2$ is again a permutation matrix. We now create every P from the identity matrix, by reordering the rows of I .

The simplest permutation matrix is $P = I$ (no exchanges). The next simplest are the row exchanges P_{ij} . Those are constructed by exchanging two rows i and j of I . Other permutations reorder more rows. By doing all possible row exchanges to I , we get all possible permutation matrices:

DEFINITION A permutation matrix P has the rows of the identity I in any order.

Example 5 There are six 3 by 3 permutation matrices. Here they are without the zeros:

$$\begin{array}{lll} I = \begin{bmatrix} 1 & & \\ & 1 & \\ & & 1 \end{bmatrix} & P_{21} = \begin{bmatrix} & 1 & \\ 1 & & \\ & & 1 \end{bmatrix} & P_{32}P_{21} = \begin{bmatrix} & 1 & \\ & & 1 \\ 1 & & \end{bmatrix} \\ P_{31} = \begin{bmatrix} & & 1 \\ & 1 & \\ 1 & & \end{bmatrix} & P_{32} = \begin{bmatrix} 1 & & \\ & & 1 \\ & 1 & \end{bmatrix} & P_{21}P_{32} = \begin{bmatrix} & & 1 \\ 1 & & \\ & 1 & \end{bmatrix}. \end{array}$$

There are $n!$ permutation matrices of order n . The symbol $n!$ means “ n factorial,” the product of the numbers $(1)(2)\cdots(n)$. Thus $3! = (1)(2)(3)$ which is 6. There will be 24 permutation matrices of order $n = 4$. And 120 permutations of order 5.

There are only two permutation matrices of order 2, namely $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ and $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$.

Important: P^{-1} is also a permutation matrix. Among the six 3 by 3 P 's displayed above, the four matrices on the left are their own inverses. The two matrices on the right are inverses of each other. In all cases, a single row exchange is its own inverse. If we repeat the exchange we are back to I . But for $P_{32}P_{21}$, the inverses go in opposite order as always. The inverse is $P_{21}P_{32}$.

More important: P^{-1} is always the same as P^T . The two matrices on the right are transposes—and inverses—of each other. When we multiply PP^T , the “1” in the first row of P hits the “1” in the first column of P^T (since the first row of P is the first column of P^T). It misses the ones in all the other columns. So $PP^T = I$.

Another proof of $P^T = P^{-1}$ looks at P as a product of row exchanges. Every row exchange is its own transpose and its own inverse. P^T and P^{-1} both come from the product of row exchanges in reverse order. So P^T and P^{-1} are the same.

Symmetric matrices led to $A = LDL^T$. Now permutations lead to $PA = LU$.

The $PA = LU$ Factorization with Row Exchanges

We sure hope you remember $A = LU$. It started with $A = (E_{21}^{-1} \cdots E_{ij}^{-1} \cdots)U$. Every elimination step was carried out by an E_{ij} and it was inverted by E_{ij}^{-1} . Those inverses were compressed into one matrix L , bringing U back to A . The lower triangular L has 1's on the diagonal, and the result is $A = LU$.

This is a great factorization, but it doesn't always work. Sometimes row exchanges are needed to produce pivots. Then $A = (E^{-1} \cdots P^{-1} \cdots E^{-1} \cdots P^{-1} \cdots)U$. Every row exchange is carried out by a P_{ij} and inverted by that P_{ij} . We now compress those row exchanges into a *single permutation matrix* P . This gives a factorization for every invertible matrix A —which we naturally want.

The main question is where to collect the P_{ij} 's. There are two good possibilities—do all the exchanges before elimination, or do them after the E_{ij} 's. The first way gives $PA = LU$. The second way has a permutation matrix P_1 in the middle.

1. The row exchanges can be done *in advance*. Their product P puts the rows of A in the right order, so that no exchanges are needed for PA . **Then $PA = LU$.**
2. If we hold row exchanges until *after elimination*, the pivot rows are in a strange order. P_1 puts them in the correct triangular order in U_1 . **Then $A = L_1 P_1 U_1$.**

$PA = LU$ is constantly used in all computing (and in MATLAB). **We will concentrate on this form.** Most numerical analysts have never seen the other form.

The factorization $A = L_1 P_1 U_1$ might be more elegant. If we mention both, it is because the difference is not well known. Probably you will not spend a long time on either one. Please don't. The most important case has $P = I$, when A equals LU with no exchanges.

For this matrix A , exchange rows 1 and 2 to put the first pivot in its usual place. Then go through elimination on PA :

$$\begin{array}{c} \begin{bmatrix} 0 & 1 & 1 \\ 1 & 2 & 1 \\ 2 & 7 & 9 \end{bmatrix} \\ A \end{array} \rightarrow \begin{array}{c} \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 2 & 7 & 9 \end{bmatrix} \\ PA \end{array} \rightarrow \begin{array}{c} \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 3 & 7 \end{bmatrix} \\ \ell_{31} = 2 \end{array} \rightarrow \begin{array}{c} \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 4 \end{bmatrix} \\ \ell_{32} = 3 \end{array}.$$

The matrix PA has its rows in good order, and it factors as usual into LU :

$$P = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad PA = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 4 \end{bmatrix} = LU. \quad (8)$$

We started with A and ended with U . *The only requirement is invertibility of A .*

If A is invertible, a permutation P will put its rows in the right order to factor $PA = LU$. There must be a full set of pivots after row exchanges for A to be invertible.

In MATLAB, $A([r \ k], :) = A([k \ r], :)$ exchanges row k with row r below it (where the k th pivot has been found). Then the `lu` code updates L and P and the sign of P :

$$\begin{array}{ll} \text{This is part of} & A([r \ k], :) = A([k \ r], :); \\ [L, U, P] = \text{lu}(A) & L([r \ k], 1 : k - 1) = L([k \ r], 1 : k - 1); \\ & P([r \ k], :) = P([k \ r], :); \\ & \text{sign} = -\text{sign} \end{array}$$

The “**sign**” of P tells whether the number of row exchanges is even ($\text{sign} = +1$). An odd number of row exchanges will produce $\text{sign} = -1$. At the start, P is I and $\text{sign} = +1$. When there is a row exchange, the sign is reversed. The final value of sign is the **determinant of P** and it does not depend on the order of the row exchanges.

For PA we get back to the familiar LU . This is the usual factorization. In reality, `lu`(A) often does not use the first available pivot. Mathematically we accept a small pivot—anything but zero. It is better if the computer looks down the column for the largest pivot. (Section 9.1 explains why this “**partial pivoting**” reduces the roundoff error.) Then P may contain row exchanges that are not algebraically necessary. Still $PA = LU$.

Our advice is to understand permutations but let the computer do the work. Calculations of $A = LU$ are enough to do by hand, without P . The Teaching Code `splu`(A) factors $PA = LU$ and `splv`(A, b) solves $Ax = b$ for any invertible A . The program `splu` stops if no pivot can be found in column k . Then A is not invertible.

■ REVIEW OF THE KEY IDEAS ■

1. The transpose puts the rows of A into the columns of A^T . Then $(A^T)_{ij} = A_{ji}$.
2. The transpose of AB is $B^T A^T$. The transpose of A^{-1} is the inverse of A^T .
3. The dot product is $x \cdot y = x^T y$. Then $(Ax)^T y$ equals the dot product $x^T (A^T y)$.
4. When A is symmetric ($A^T = A$), its LDU factorization is symmetric: $A = LDL^T$.
5. A permutation matrix P has a 1 in each row and column, and $P^T = P^{-1}$.
6. There are $n!$ permutation matrices of size n . *Half even, half odd.*
7. If A is invertible then a permutation P will reorder its rows for $PA = LU$.

■ WORKED EXAMPLES ■

2.7 A Applying the permutation P to the rows of A destroys its symmetry:

$$P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \quad A = \begin{bmatrix} 1 & 4 & 5 \\ 4 & 2 & 6 \\ 5 & 6 & 3 \end{bmatrix} \quad PA = \begin{bmatrix} 4 & 2 & 6 \\ 5 & 6 & 3 \\ 1 & 4 & 5 \end{bmatrix}$$

What permutation Q applied to the *columns* of PA will recover symmetry in PAQ ? The numbers 1, 2, 3 must come back to the main diagonal (not necessarily in order). Show that Q is P^T , so that symmetry is saved by $PAQ = PAP^T$.

Solution To recover symmetry and put “2” back on the diagonal, column 2 of PA must move to column 1. Column 3 of PA (containing “3”) must move to column 2. Then the “1” moves to the 3, 3 position. The matrix that permutes columns is Q :

$$PA = \begin{bmatrix} 4 & 2 & 6 \\ 5 & 6 & 3 \\ 1 & 4 & 5 \end{bmatrix} \quad Q = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \quad PAQ = \begin{bmatrix} 2 & 6 & 4 \\ 6 & 3 & 5 \\ 4 & 5 & 1 \end{bmatrix} \text{ is symmetric.}$$

The matrix Q is P^T . This choice always recovers symmetry, because PAP^T is guaranteed to be symmetric. (Its transpose is again PAP^T .) The matrix Q is also P^{-1} , because the inverse of every permutation matrix is its transpose.

If D is a diagonal matrix, we are finding that PDP^T is also diagonal. When P moves row 1 down to row 3, P^T on the right will move column 1 to column 3. The (1, 1) entry moves down to (3, 1) and over to (3, 3).

2.7 B Find the symmetric factorization $A = LDL^T$ for the matrix A above. Is this A invertible? Find also the $PQ = LU$ factorization for Q , which needs row exchanges.

Solution To factor A into LDL^T we eliminate below the pivots:

$$A = \begin{bmatrix} 1 & 4 & 5 \\ 4 & 2 & 6 \\ 5 & 6 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 4 & 5 \\ 0 & -14 & -14 \\ 0 & -14 & -22 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 4 & 5 \\ 0 & -14 & -14 \\ 0 & 0 & -8 \end{bmatrix} = U.$$

The multipliers were $\ell_{21} = 4$ and $\ell_{31} = 5$ and $\ell_{32} = 1$. The pivots 1, -14, -8 go into D . When we divide the rows of U by those pivots, L^T should appear:

$$\begin{array}{l} \text{Symmetric} \\ \text{factorization} \\ \text{when } A = A^T \end{array} \quad A = LDL^T = \begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 5 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & & \\ & -14 & \\ & & -8 \end{bmatrix} \begin{bmatrix} 1 & 4 & 5 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}.$$

This matrix A is invertible because it has three pivots. Its inverse is $(L^T)^{-1}D^{-1}L^{-1}$ and A^{-1} is also symmetric. The numbers 14 and 8 will turn up in the denominators of A^{-1} . The “determinant” of A is the product of the pivots $(1)(-14)(-8) = 112$.

Any permutation matrix Q is invertible. Here elimination needs two row exchanges:

$$Q = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \xrightarrow[\substack{\text{rows} \\ 1 \leftrightarrow 2}]{\text{rows}} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \xrightarrow[\substack{\text{rows} \\ 2 \leftrightarrow 3}]{\text{rows}} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I.$$

With $A = Q$, the $PQ = (L)(U)$ factorization is the same as $Q^{-1}Q = (I)(I)$.

2.7 C For a rectangular A , this *saddle-point matrix* S is symmetric and important:

$$\begin{array}{l} \text{Block matrix} \\ \text{from least squares} \end{array} \quad S = \begin{bmatrix} I & A \\ A^T & 0 \end{bmatrix} = S^T \text{ has size } m + n.$$

Apply block elimination to find a **block factorization** $S = LDL^T$. Then test invertibility:

$$S \text{ is invertible} \iff A^T A \text{ is invertible} \iff Ax \neq 0 \text{ whenever } x \neq 0$$

Solution The first block pivot is I . The matrix to multiply row 1 is certainly A^T :

$$\text{Block elimination} \quad S = \begin{bmatrix} I & A \\ A^T & 0 \end{bmatrix} \text{ goes to } \begin{bmatrix} I & A \\ 0 & -A^T A \end{bmatrix}. \text{ This is } U.$$

The block pivot matrix D contains I and $-A^T A$. Then L and L^T contain A^T and A :

$$\text{Block factorization} \quad S = LDL^T = \begin{bmatrix} I & 0 \\ A^T & I \end{bmatrix} \begin{bmatrix} I & 0 \\ 0 & -A^T A \end{bmatrix} \begin{bmatrix} I & A \\ 0 & I \end{bmatrix}.$$

L is certainly invertible, with diagonal 1's from I . The inverse of the middle matrix involves $(A^T A)^{-1}$. Section 4.2 answers a key question about the matrix $A^T A$:

When is $A^T A$ invertible? *Answer: A must have independent columns.*
Then $Ax = 0$ only if $x = 0$. Otherwise $Ax = 0$ will lead to $A^T Ax = 0$.

Problem Set 2.7

Questions 1–7 are about the rules for transpose matrices.

- 1 Find A^T and A^{-1} and $(A^{-1})^T$ and $(A^T)^{-1}$ for

$$A = \begin{bmatrix} 1 & 0 \\ 9 & 3 \end{bmatrix} \quad \text{and also} \quad A = \begin{bmatrix} 1 & c \\ c & 0 \end{bmatrix}.$$

- 2 Verify that $(AB)^T$ equals $B^T A^T$ but those are different from $A^T B^T$:

$$A = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} \quad AB = \begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix}.$$

In case $AB = BA$ (not generally true!) how do you prove that $B^T A^T = A^T B^T$?

- 3 (a) The matrix $((AB)^{-1})^T$ comes from $(A^{-1})^T$ and $(B^{-1})^T$. *In what order?*
 (b) If U is upper triangular then $(U^{-1})^T$ is _____ triangular.
- 4 Show that $A^2 = 0$ is possible but $A^T A = 0$ is not possible (unless $A =$ zero matrix).
- 5 (a) The row vector x^T times A times the column y produces what number?

$$x^T A y = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \underline{\hspace{2cm}}.$$

- (b) This is the row $x^T A =$ _____ times the column $y = (0, 1, 0)$.
 (c) This is the row $x^T = [0 \ 1]$ times the column $Ay =$ _____.
- 6 The transpose of a block matrix $M = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$ is $M^T =$ _____. Test an example. Under what conditions on A, B, C, D is the block matrix symmetric?
- 7 True or false:
- (a) The block matrix $\begin{bmatrix} 0 & A \\ A & 0 \end{bmatrix}$ is automatically symmetric.
 (b) If A and B are symmetric then their product AB is symmetric.
 (c) If A is not symmetric then A^{-1} is not symmetric.
 (d) When A, B, C are symmetric, the transpose of ABC is CBA .

Questions 8–15 are about permutation matrices.

- 8 Why are there $n!$ permutation matrices of order n ?
- 9 If P_1 and P_2 are permutation matrices, so is $P_1 P_2$. This still has the rows of I in some order. Give examples with $P_1 P_2 \neq P_2 P_1$ and $P_3 P_4 = P_4 P_3$.
- 10 There are 12 “even” permutations of $(1, 2, 3, 4)$, with an *even number of exchanges*. Two of them are $(1, 2, 3, 4)$ with no exchanges and $(4, 3, 2, 1)$ with two exchanges. List the other ten. Instead of writing each 4 by 4 matrix, just order the numbers.
- 11 Which permutation makes PA upper triangular? Which permutations make $P_1 A P_2$ lower triangular? **Multiplying A on the right by P_2 exchanges the _____ of A .**

$$A = \begin{bmatrix} 0 & 0 & 6 \\ 1 & 2 & 3 \\ 0 & 4 & 5 \end{bmatrix}.$$

- 12 Explain why the dot product of x and y equals the dot product of Px and Py . Then from $(Px)^T(Py) = x^T y$ deduce that $P^T P = I$ for any permutation. With $x = (1, 2, 3)$ and $y = (1, 4, 2)$ choose P to show that $Px \cdot y$ is not always $x \cdot Py$.
- 13 (a) Find a 3 by 3 permutation matrix with $P^3 = I$ (but not $P = I$).
 (b) Find a 4 by 4 permutation $\hat{P}$ with $\hat{P}^4 \neq I$.

- 14 If P has 1's on the antidiagonal from $(1, n)$ to $(n, 1)$, describe PAP . Note $P = P^T$.
- 15 All row exchange matrices are symmetric: $P^T = P$. Then $P^T P = I$ becomes $P^2 = I$. Other permutation matrices may or may not be symmetric.
- (a) If P sends row 1 to row 4, then P^T sends row _____ to row _____.
When $P^T = P$ the row exchanges come in pairs with no overlap.
- (b) Find a 4 by 4 example with $P^T = P$ that moves all four rows.

Questions 16–21 are about symmetric matrices and their factorizations.

- 16 If $A = A^T$ and $B = B^T$, which of these matrices are certainly symmetric?
- (a) $A^2 - B^2$ (b) $(A + B)(A - B)$ (c) ABA (d) $ABAB$.
- 17 Find 2 by 2 symmetric matrices $A = A^T$ with these properties:
- (a) A is not invertible.
(b) A is invertible but cannot be factored into LU (row exchanges needed).
(c) A can be factored into LDL^T but not into LL^T (because of negative D).
- 18 (a) How many entries of A can be chosen independently, if $A = A^T$ is 5 by 5?
(b) How do L and D (still 5 by 5) give the same number of choices in LDL^T ?
(c) How many entries can be chosen if A is *skew-symmetric*? ($A^T = -A$).
- 19 Suppose R is rectangular (m by n) and A is symmetric (m by m).
- (a) Transpose $R^T A R$ to show its symmetry. What shape is this matrix?
(b) Show why $R^T R$ has no negative numbers on its diagonal.
- 20 Factor these symmetric matrices into $A = LDL^T$. The pivot matrix D is diagonal:

$$A = \begin{bmatrix} 1 & 3 \\ 3 & 2 \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} 1 & b \\ b & c \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}.$$

- 21 After elimination clears out column 1 below the first pivot, find the symmetric 2 by 2 matrix that appears in the lower right corner:

$$\text{Start from } A = \begin{bmatrix} 2 & 4 & 8 \\ 4 & 3 & 9 \\ 8 & 9 & 0 \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} 1 & b & c \\ b & d & e \\ c & e & f \end{bmatrix}.$$

Questions 22–24 are about the factorizations $PA = LU$ and $A = L_1 P_1 U_1$.

- 22 Find the $PA = LU$ factorizations (and check them) for

$$A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 2 & 3 & 4 \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 4 & 1 \\ 1 & 1 & 1 \end{bmatrix}.$$

- 23 Find a 4 by 4 permutation matrix (call it A) that needs 3 row exchanges to reach the end of elimination. For this matrix, what are its factors P , L , and U ?
- 24 Factor the following matrix into $PA = LU$. Factor it also into $A = L_1 P_1 U_1$ (hold the exchange of row 3 until 3 times row 1 is subtracted from row 2):

$$A = \begin{bmatrix} 0 & 1 & 2 \\ 0 & 3 & 8 \\ 2 & 1 & 1 \end{bmatrix}.$$

- 25 Extend the **slu** code in Section 2.6 to a code **splu** that factors PA into LU .
- 26 Prove that the identity matrix cannot be the product of three row exchanges (or five). It can be the product of two exchanges (or four).
- 27 (a) Choose E_{21} to remove the 3 below the first pivot. Then multiply $E_{21} A E_{21}^T$ to remove both 3's:

$$A = \begin{bmatrix} 1 & 3 & 0 \\ 3 & 11 & 4 \\ 0 & 4 & 9 \end{bmatrix} \quad \text{is going toward} \quad D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

- (b) Choose E_{32} to remove the 4 below the second pivot. Then A is reduced to D by $E_{32} E_{21} A E_{21}^T E_{32}^T = D$. Invert the E 's to find L in $A = LDL^T$.
- 28 If every row of a 4 by 4 matrix contains the numbers 0, 1, 2, 3 in some order, can the matrix be symmetric?
- 29 Prove that no reordering of rows and reordering of columns can transpose a typical matrix. (Watch the diagonal entries.)

The next three questions are about applications of the identity $(Ax)^T y = x^T (A^T y)$.

- 30 Wires go between Boston, Chicago, and Seattle. Those cities are at voltages x_B , x_C , x_S . With unit resistances between cities, the currents between cities are in y :

$$y = Ax \quad \text{is} \quad \begin{bmatrix} y_{BC} \\ y_{CS} \\ y_{BS} \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} x_B \\ x_C \\ x_S \end{bmatrix}.$$

- (a) Find the total currents $A^T y$ out of the three cities.
- (b) Verify that $(Ax)^T y$ agrees with $x^T (A^T y)$ —six terms in both.

- 31 Producing x_1 trucks and x_2 planes needs $x_1 + 50x_2$ tons of steel, $40x_1 + 1000x_2$ pounds of rubber, and $2x_1 + 50x_2$ months of labor. If the unit costs y_1, y_2, y_3 are \$700 per ton, \$3 per pound, and \$3000 per month, what are the values of one truck and one plane? Those are the components of $A^T y$.
- 32 Ax gives the amounts of steel, rubber, and labor to produce x in Problem 31. Find A . Then $Ax \cdot y$ is the _____ of inputs while $x \cdot A^T y$ is the value of _____.
- 33 The matrix P that multiplies (x, y, z) to give (z, x, y) is also a rotation matrix. Find P and P^3 . The rotation axis $a = (1, 1, 1)$ doesn't move, it equals Pa . What is the angle of rotation from $v = (2, 3, -5)$ to $Pv = (-5, 2, 3)$?
- 34 Write $A = \begin{bmatrix} 1 & 2 \\ 4 & 9 \end{bmatrix}$ as the product EH of an elementary row operation matrix E and a symmetric matrix H .
- 35 Here is a new factorization of A into *triangular* (with 1's) *times symmetric*:

$$\text{Start from } A = LDU. \text{ Then } A = L(U^T)^{-1} \text{ times } U^T D U.$$

Why is $L(U^T)^{-1}$ triangular? Its diagonal is all 1's. Why is $U^T D U$ symmetric?

- 36 A *group* of matrices includes AB and A^{-1} if it includes A and B . "Products and inverses stay in the group." Which of these sets are groups?
Lower triangular matrices L with 1's on the diagonal, symmetric matrices S , positive matrices M , diagonal invertible matrices D , permutation matrices P , matrices with $Q^T = Q^{-1}$. *Invent two more matrix groups.*

Challenge Problems

- 37 A square **northwest matrix** B is zero in the southeast corner, below the antidiagonal that connects $(1, n)$ to $(n, 1)$. Will B^T and B^2 be northwest matrices? Will B^{-1} be northwest or southeast? What is the shape of $BC = \text{northwest times southeast}$?
- 38 If you take powers of a permutation matrix, why is some P^k eventually equal to I ? Find a 5 by 5 permutation P so that the smallest power to equal I is P^6 .
- 39 (a) Write down any 3 by 3 matrix A . Split A into $B + C$ where $B = B^T$ is symmetric and $C = -C^T$ is anti-symmetric.
(b) Find formulas for B and C involving A and A^T . We want $A = B + C$ with $B = B^T$ and $C = -C^T$.
- 40 Suppose Q^T equals Q^{-1} (transpose equals inverse, so $Q^T Q = I$).
(a) Show that the columns $q_1, \dots, q_n$ are unit vectors: $\|q_i\|^2 = 1$.
(b) Show that every two columns of Q are perpendicular: $q_1^T q_2 = 0$.
(c) Find a 2 by 2 example with first entry $q_{11} = \cos \theta$.

Chapter 3

Vector Spaces and Subspaces

3.1 Spaces of Vectors

To a newcomer, matrix calculations involve a lot of numbers. To you, they involve vectors. The columns of Ax and AB are linear combinations of n vectors—the columns of A . This chapter moves from numbers and vectors to a third level of understanding (the highest level). Instead of individual columns, we look at “spaces” of vectors. Without seeing *vector spaces* and especially their *subspaces*, you haven’t understood everything about $Ax = b$.

Since this chapter goes a little deeper, it may seem a little harder. That is natural. We are looking inside the calculations, to find the mathematics. The author’s job is to make it clear. The chapter ends with the “*Fundamental Theorem of Linear Algebra*”.

We begin with the most important vector spaces. They are denoted by $\mathbf{R}^1$, $\mathbf{R}^2$, $\mathbf{R}^3$, $\mathbf{R}^4$, Each space $\mathbf{R}^n$ consists of a whole collection of vectors. $\mathbf{R}^5$ contains all column vectors with five components. This is called “5-dimensional space”.

DEFINITION *The space $\mathbf{R}^n$ consists of all column vectors v with n components.*

The components of v are real numbers, which is the reason for the letter $\mathbf{R}$. A vector whose n components are complex numbers lies in the space $\mathbf{C}^n$.

The vector space $\mathbf{R}^2$ is represented by the usual xy plane. Each vector v in $\mathbf{R}^2$ has two components. The word “*space*” asks us to think of all those vectors—the whole plane. Each vector gives the x and y coordinates of a point in the plane: $v = (x, y)$.

Similarly the vectors in $\mathbf{R}^3$ correspond to points (x, y, z) in three-dimensional space. The one-dimensional space $\mathbf{R}^1$ is a line (like the x axis). As before, we print vectors as a column between brackets, or along a line using commas and parentheses:

$$\begin{bmatrix} 4 \\ \pi \end{bmatrix} \text{ is in } \mathbf{R}^2, \quad (1, 1, 0, 1, 1) \text{ is in } \mathbf{R}^5, \quad \begin{bmatrix} 1+i \\ 1-i \end{bmatrix} \text{ is in } \mathbf{C}^2.$$

The great thing about linear algebra is that it deals easily with five-dimensional space. We don’t draw the vectors, we just need the five numbers (or n numbers).

To multiply v by 7, multiply every component by 7. Here 7 is a “scalar”. To add vectors in $\mathbf{R}^5$, add them a component at a time. The two essential vector operations go on *inside the vector space*, and they produce *linear combinations*:

We can add any vectors in $\mathbf{R}^n$, and we can multiply any vector v by any scalar c .

“Inside the vector space” means that *the result stays in the space*. If v is the vector in $\mathbf{R}^4$ with components 1, 0, 0, 1, then $2v$ is the vector in $\mathbf{R}^4$ with components 2, 0, 0, 2. (In this case 2 is the scalar.) A whole series of properties can be verified in $\mathbf{R}^n$. The commutative law is $v + w = w + v$; the distributive law is $c(v + w) = cv + cw$. There is a unique “zero vector” satisfying $0 + v = v$. Those are three of the eight conditions listed at the start of the problem set.

These eight conditions are required of every vector space. There are vectors other than column vectors, and vector spaces other than $\mathbf{R}^n$, and all vector spaces have to obey the eight reasonable rules.

A real vector space is a set of “vectors” together with rules for vector addition and for multiplication by real numbers. The addition and the multiplication must produce vectors that are in the space. And the eight conditions must be satisfied (which is usually no problem). Here are three vector spaces other than $\mathbf{R}^n$:

- M** The vector space of *all real 2 by 2 matrices*.
- F** The vector space of *all real functions $f(x)$* .
- Z** The vector space that consists only of a *zero vector*.

In **M** the “vectors” are really matrices. In **F** the vectors are functions. In **Z** the only addition is $0 + 0 = 0$. In each case we can add: matrices to matrices, functions to functions, zero vector to zero vector. We can multiply a matrix by 4 or a function by 4 or the zero vector by 4. The result is still in **M** or **F** or **Z**. The eight conditions are all easily checked.

The function space **F** is infinite-dimensional. A smaller function space is **P**, or **P_n**, containing all polynomials $a_0 + a_1x + \cdots + a_nx^n$ of degree n .

The space **Z** is zero-dimensional (by any reasonable definition of dimension). It is the smallest possible vector space. We hesitate to call it $\mathbf{R}^0$, which means no components—you might think there was no vector. The vector space **Z** contains exactly *one vector* (zero). No space can do without that zero vector. Each space has its own zero vector—the zero matrix, the zero function, the vector (0, 0, 0) in $\mathbf{R}^3$.

Subspaces

At different times, we will ask you to think of matrices and functions as vectors. But at all times, the vectors that we need most are ordinary column vectors. They are vectors with n components—but *maybe not all* of the vectors with n components. There are important vector spaces *inside* $\mathbf{R}^n$. Those are *subspaces* of $\mathbf{R}^n$.

Start with the usual three-dimensional space $\mathbf{R}^3$. Choose a plane through the origin (0, 0, 0). *That plane is a vector space in its own right*. If we add two vectors in the plane, their sum is in the plane. If we multiply an in-plane vector by 2 or -5 , it is still in the plane.

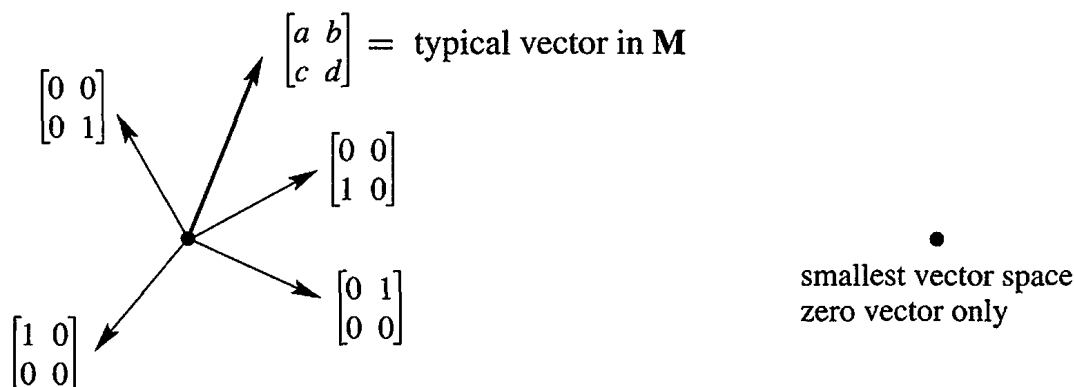


Figure 3.1: “Four-dimensional” matrix space M . The “zero-dimensional” space Z .

A plane in three-dimensional space is not $\mathbf{R}^2$ (even if it looks like $\mathbf{R}^2$). The vectors have three components and they belong to $\mathbf{R}^3$. The plane is a vector space *inside* $\mathbf{R}^3$.

This illustrates one of the most fundamental ideas in linear algebra. The plane going through $(0, 0, 0)$ is a *subspace* of the full vector space $\mathbf{R}^3$.

DEFINITION A *subspace* of a vector space is a set of vectors (including $\mathbf{0}$) that satisfies two requirements: *If v and w are vectors in the subspace and c is any scalar, then*

(i) $v + w$ is in the subspace

(ii) cv is in the subspace.

In other words, the set of vectors is “closed” under addition $v + w$ and multiplication cv (and cw). Those operations leave us in the subspace. We can also subtract, because $-w$ is in the subspace and its sum with v is $v - w$. In short, *all linear combinations stay in the subspace*.

All these operations follow the rules of the host space, so the eight required conditions are automatic. We just have to check the requirements for a subspace, so that we can take linear combinations.

First fact: *Every subspace contains the zero vector*. The plane in $\mathbf{R}^3$ has to go through $(0, 0, 0)$. We mention this separately, for extra emphasis, but it follows directly from rule (ii). Choose $c = 0$, and the rule requires $0v$ to be in the subspace.

Planes that don’t contain the origin fail those tests. When v is on such a plane, $-v$ and $0v$ are *not* on the plane. A plane that misses the origin is not a subspace.

Lines through the origin are also subspaces. When we multiply by 5, or add two vectors on the line, we stay on the line. But the line must go through $(0, 0, 0)$.

Another subspace is all of $\mathbf{R}^3$. The whole space is a subspace (*of itself*). Here is a list of all the possible subspaces of $\mathbf{R}^3$:

- | | |
|-----------------------------------|------------------------------------|
| (L) Any line through $(0, 0, 0)$ | ($\mathbf{R}^3$) The whole space |
| (P) Any plane through $(0, 0, 0)$ | (Z) The single vector $(0, 0, 0)$ |

If we try to keep only *part* of a plane or line, the requirements for a subspace don't hold. Look at these examples in $\mathbf{R}^2$.

Example 1 Keep only the vectors (x, y) whose components are positive or zero (this is a quarter-plane). The vector $(2, 3)$ is included but $(-2, -3)$ is not. So rule (ii) is violated when we try to multiply by $c = -1$. *The quarter-plane is not a subspace.*

Example 2 Include also the vectors whose components are both negative. Now we have two quarter-planes. Requirement (ii) is satisfied; we can multiply by any c . But rule (i) now fails. The sum of $v = (2, 3)$ and $w = (-3, -2)$ is $(-1, 1)$, which is outside the quarter-planes. *Two quarter-planes don't make a subspace.*

Rules (i) and (ii) involve vector addition $v + w$ and multiplication by scalars like c and d . The rules can be combined into a single requirement—the rule for subspaces:

A subspace containing v and w must contain all linear combinations $cv + dw$.

Example 3 Inside the vector space $\mathbf{M}$ of all 2 by 2 matrices, here are two subspaces:

- (U) All upper triangular matrices $\begin{bmatrix} a & b \\ 0 & d \end{bmatrix}$ (D) All diagonal matrices $\begin{bmatrix} a & 0 \\ 0 & d \end{bmatrix}$.

Add any two matrices in U, and the sum is in U. Add diagonal matrices, and the sum is diagonal. In this case D is also a subspace of U! Of course the zero matrix is in these subspaces, when a, b , and d all equal zero.

To find a smaller subspace of diagonal matrices, we could require $a = d$. The matrices are multiples of the identity matrix I . The sum $2I + 3I$ is in this subspace, and so is 3 times $4I$. The matrices cI form a “line of matrices” inside $\mathbf{M}$ and U and D.

Is the matrix I a subspace by itself? Certainly not. Only the zero matrix is. Your mind will invent more subspaces of 2 by 2 matrices—write them down for Problem 5.

The Column Space of A

The most important subspaces are tied directly to a matrix A . We are trying to solve $Ax = b$. If A is not invertible, the system is solvable for some b and not solvable for other b . We want to describe the good right sides b —the vectors that *can* be written as A times some vector x . Those b 's form the “column space” of A .

Remember that Ax is a combination of the columns of A . To get every possible b , we use every possible x . So start with the columns of A , and *take all their linear combinations*. *This produces the column space of A . It is a vector space made up of column vectors.*

$C(A)$ contains not just the n columns of A , but all their combinations Ax .

DEFINITION The *column space* consists of *all linear combinations of the columns*. The combinations are all possible vectors Ax . They fill the column space $C(A)$.

This column space is crucial to the whole book, and here is why. *To solve $Ax = b$ is to express b as a combination of the columns*. The right side b has to be *in the column space* produced by A on the left side, or no solution!

The system $Ax = b$ is solvable if and only if b is in the column space of A .

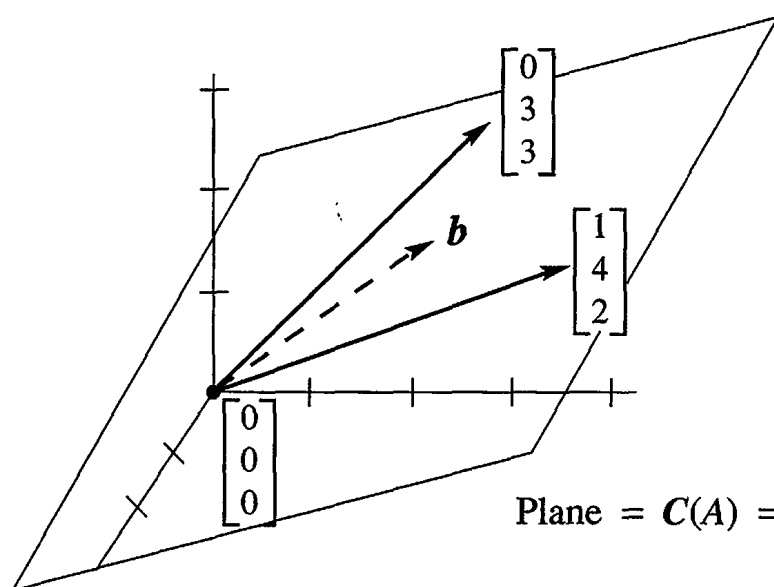
When b is in the column space, it is a combination of the columns. The coefficients in that combination give us a solution x to the system $Ax = b$.

Suppose A is an m by n matrix. Its columns have m components (not n). So the columns belong to $\mathbb{R}^m$. *The column space of A is a subspace of $\mathbb{R}^m$ (not $\mathbb{R}^n$)*. The set of all column combinations Ax satisfies rules (i) and (ii) for a subspace: When we add linear combinations or multiply by scalars, we still produce combinations of the columns. The word “subspace” is justified by *taking all linear combinations*.

Here is a 3 by 2 matrix A , whose column space is a subspace of $\mathbb{R}^3$. The column space of A is a plane in Figure 3.2.

Example 4

$$Ax \text{ is } \begin{bmatrix} 1 & 0 \\ 4 & 3 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \text{ which is } x_1 \begin{bmatrix} 1 \\ 4 \\ 2 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 3 \\ 3 \end{bmatrix}.$$



$$A = \begin{bmatrix} 1 & 0 \\ 4 & 3 \\ 2 & 3 \end{bmatrix}$$

$$b = x_1 \begin{bmatrix} 1 \\ 4 \\ 2 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 3 \\ 3 \end{bmatrix}$$

Plane = $C(A)$ = all vectors Ax

Figure 3.2: The column space $C(A)$ is a plane containing the two columns. $Ax = b$ is solvable when b is on that plane. Then b is a combination of the columns.

The column space of all combinations of the two columns *fills up a plane in $\mathbf{R}^3$* . We drew one particular $\mathbf{b}$ (a combination of the columns). This $\mathbf{b} = A\mathbf{x}$ lies on the plane. The plane has zero thickness, so most right sides $\mathbf{b}$ in $\mathbf{R}^3$ are *not* in the column space. For most $\mathbf{b}$ there is no solution to our 3 equations in 2 unknowns.

Of course $(0, 0, 0)$ is in the column space. The plane passes through the origin. There is certainly a solution to $A\mathbf{x} = \mathbf{0}$. That solution, always available, is $\mathbf{x} = \underline{\hspace{1cm}}$.

To repeat, the attainable right sides $\mathbf{b}$ are exactly the vectors in the column space. One possibility is the first column itself—take $x_1 = 1$ and $x_2 = 0$. Another combination is the second column—take $x_1 = 0$ and $x_2 = 1$. The new level of understanding is to see *all* combinations—the whole subspace is generated by those two columns.

Notation The column space of A is denoted by $C(A)$. Start with the columns and take all their linear combinations. We might get the whole $\mathbf{R}^m$ or only a subspace.

Important Instead of columns in $\mathbf{R}^m$, we could start with any set S of vectors in a vector space V . To get a subspace SS of V , we take *all combinations* of the vectors in that set:

$$\begin{aligned} S &= \text{set of vectors in } V \text{ (probably not a subspace)} \\ SS &= \text{all combinations of vectors in } S \end{aligned}$$

$$SS = \text{all } c_1\mathbf{v}_1 + \cdots + c_N\mathbf{v}_N = \text{the subspace of } V \text{ “spanned” by } S$$

When S is the set of columns, SS is the column space. When there is only one nonzero vector $\mathbf{v}$ in S , the subspace SS is the line through $\mathbf{v}$. *Always SS is the smallest subspace containing S .* This is a fundamental way to create subspaces and we will come back to it.

The subspace SS is the “span” of S , containing all combinations of vectors in S .

Example 5 Describe the column spaces (they are subspaces of $\mathbf{R}^2$) for

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 4 \end{bmatrix}.$$

Solution The column space of I is the *whole space* $\mathbf{R}^2$. Every vector is a combination of the columns of I . In vector space language, $C(I)$ is $\mathbf{R}^2$.

The column space of A is only a line. The second column $(2, 4)$ is a multiple of the first column $(1, 2)$. Those vectors are different, but our eye is on vector *spaces*. The column space contains $(1, 2)$ and $(2, 4)$ and all other vectors $(c, 2c)$ along that line. The equation $A\mathbf{x} = \mathbf{b}$ is only solvable when $\mathbf{b}$ is on the line.

For the third matrix (with three columns) the column space $C(B)$ is all of $\mathbf{R}^2$. Every $\mathbf{b}$ is attainable. The vector $\mathbf{b} = (5, 4)$ is column 2 plus column 3, so $\mathbf{x}$ can be $(0, 1, 1)$. The same vector $(5, 4)$ is also $2(\text{column } 1) + \text{column } 3$, so another possible $\mathbf{x}$ is $(2, 0, 1)$. This matrix has the same column space as I —any $\mathbf{b}$ is allowed. But now $\mathbf{x}$ has extra components and there are more solutions—more combinations that give $\mathbf{b}$.

The next section creates a vector space $N(A)$, to describe all the solutions of $A\mathbf{x} = \mathbf{0}$. This section created the column space $C(A)$, to describe all the attainable right sides $\mathbf{b}$.

■ REVIEW OF THE KEY IDEAS ■

1. $\mathbf{R}^n$ contains all column vectors with n real components.
2. $\mathbf{M}$ (2 by 2 matrices) and $\mathbf{F}$ (functions) and $\mathbf{Z}$ (zero vector alone) are vector spaces.
3. A subspace containing $\mathbf{v}$ and $\mathbf{w}$ must contain all their combinations $c\mathbf{v} + d\mathbf{w}$.
4. The combinations of the columns of A form the *column space* $C(A)$. Then the column space is “spanned” by the columns.
5. $A\mathbf{x} = \mathbf{b}$ has a solution exactly when $\mathbf{b}$ is in the column space of A .

■ WORKED EXAMPLES ■

3.1 A We are given three different vectors $\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3$. Construct a matrix so that the equations $A\mathbf{x} = \mathbf{b}_1$ and $A\mathbf{x} = \mathbf{b}_2$ are solvable, but $A\mathbf{x} = \mathbf{b}_3$ is not solvable. How can you decide if this is possible? How could you construct A ?

Solution We want to have $\mathbf{b}_1$ and $\mathbf{b}_2$ in the column space of A . Then $A\mathbf{x} = \mathbf{b}_1$ and $A\mathbf{x} = \mathbf{b}_2$ will be solvable. *The quickest way is to make $\mathbf{b}_1$ and $\mathbf{b}_2$ the two columns of A .* Then the solutions are $\mathbf{x} = (1, 0)$ and $\mathbf{x} = (0, 1)$.

Also, we don't want $A\mathbf{x} = \mathbf{b}_3$ to be solvable. So don't make the column space any larger! Keeping only the columns of $\mathbf{b}_1$ and $\mathbf{b}_2$, the question is:

$$\text{Is } A\mathbf{x} = \begin{bmatrix} \mathbf{b}_1 & \mathbf{b}_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \mathbf{b}_3 \text{ solvable?} \quad \text{Is } \mathbf{b}_3 \text{ a combination of } \mathbf{b}_1 \text{ and } \mathbf{b}_2?$$

If the answer is *no*, we have the desired matrix A . If the answer is *yes*, then it is *not possible* to construct A . When the column space contains $\mathbf{b}_1$ and $\mathbf{b}_2$, it will have to contain all their linear combinations. So $\mathbf{b}_3$ would necessarily be in that column space and $A\mathbf{x} = \mathbf{b}_3$ would necessarily be solvable.

3.1 B Describe a subspace $\mathbf{S}$ of each vector space $\mathbf{V}$, and then a subspace $\mathbf{SS}$ of $\mathbf{S}$.

$\mathbf{V}_1$ = all combinations of $(1, 1, 0, 0)$ and $(1, 1, 1, 0)$ and $(1, 1, 1, 1)$

$\mathbf{V}_2$ = all vectors perpendicular to $\mathbf{u} = (1, 2, 1)$, so $\mathbf{u} \cdot \mathbf{v} = 0$

$\mathbf{V}_3$ = all symmetric 2 by 2 matrices (a subspace of $\mathbf{M}$)

$\mathbf{V}_4$ = all solutions to the equation $d^4 y / dx^4 = 0$ (a subspace of $\mathbf{F}$)

Describe each $\mathbf{V}$ two ways: *All combinations of , all solutions of the equations*

Solution V_1 starts with three vectors. A subspace S comes from all combinations of the first two vectors $(1, 1, 0, 0)$ and $(1, 1, 1, 0)$. A subspace SS of S comes from all multiples $(c, c, 0, 0)$ of the first vector. So many possibilities.

A subspace S of V_2 is the line through $(1, -1, 1)$. This line is perpendicular to u . The vector $x = (0, 0, 0)$ is in S and all its multiples cx give the smallest subspace $SS = Z$.

The diagonal matrices are a subspace S of the symmetric matrices. The multiples cI are a subspace SS of the diagonal matrices.

V_4 contains all cubic polynomials $y = a + bx + cx^2 + dx^3$, with $d^4y/dx^4 = 0$. The quadratic polynomials give a subspace S . The linear polynomials are one choice of SS . The constants could be SSS .

In all four parts we could take $S = V$ itself, and $SS =$ the zero subspace Z .

Each V can be described as *all combinations of* and as *all solutions of*:

$V_1 =$ all combinations of the 3 vectors $V_1 =$ all solutions of $v_1 - v_2 = 0$

$V_2 =$ all combinations of $(1, 0, -1)$ and $(1, -1, 1)$ are solutions of $u \cdot v = 0$.

$V_3 =$ all combinations of $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$. $V_3 =$ all solutions $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ of $b = c$

$V_4 =$ all combinations of $1, x, x^2, x^3$ $V_4 =$ all solutions to $d^4y/dx^4 = 0$.

Problem Set 3.1

The first problems 1–8 are about vector spaces in general. The vectors in those spaces are not necessarily column vectors. In the definition of a *vector space*, vector addition $x + y$ and scalar multiplication cx must obey the following eight rules:

- (1) $x + y = y + x$
- (2) $x + (y + z) = (x + y) + z$
- (3) There is a unique “zero vector” such that $x + 0 = x$ for all x
- (4) For each x there is a unique vector $-x$ such that $x + (-x) = 0$
- (5) 1 times x equals x
- (6) $(c_1c_2)x = c_1(c_2x)$
- (7) $c(x + y) = cx + cy$
- (8) $(c_1 + c_2)x = c_1x + c_2x$.

- 1 Suppose $(x_1, x_2) + (y_1, y_2)$ is defined to be $(x_1 + y_2, x_2 + y_1)$. With the usual multiplication $cx = (cx_1, cx_2)$, which of the eight conditions are not satisfied?
- 2 Suppose the multiplication cx is defined to produce $(cx_1, 0)$ instead of (cx_1, cx_2) . With the usual addition in $\mathbf{R}^2$, are the eight conditions satisfied?

- 3 (a) Which rules are broken if we keep only the positive numbers $x > 0$ in $\mathbf{R}^1$? Every c must be allowed. The half-line is not a subspace.
 (b) The positive numbers with $x + y$ and cx redefined to equal the usual xy and x^c do satisfy the eight rules. Test rule 7 when $c = 3, x = 2, y = 1$. (Then $x + y = 2$ and $cx = 8$.) Which number acts as the “zero vector”?
- 4 The matrix $A = \begin{bmatrix} 2 & -2 \\ 2 & -2 \end{bmatrix}$ is a “vector” in the space $\mathbf{M}$ of all 2 by 2 matrices. Write down the zero vector in this space, the vector $\frac{1}{2}A$, and the vector $-A$. What matrices are in the smallest subspace containing A ?
- 5 (a) Describe a subspace of $\mathbf{M}$ that contains $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ but not $B = \begin{bmatrix} 0 & 0 \\ 0 & -1 \end{bmatrix}$.
 (b) If a subspace of $\mathbf{M}$ contains A and B , must it contain I ?
 (c) Describe a subspace of $\mathbf{M}$ that contains no nonzero diagonal matrices.
- 6 The functions $f(x) = x^2$ and $g(x) = 5x$ are “vectors” in $\mathbf{F}$. This is the vector space of all real functions. (The functions are defined for $-\infty < x < \infty$.) The combination $3f(x) - 4g(x)$ is the function $h(x) = \underline{\hspace{2cm}}$.
- 7 Which rule is broken if multiplying $f(x)$ by c gives the function $f(cx)$? Keep the usual addition $f(x) + g(x)$.
- 8 If the sum of the “vectors” $f(x)$ and $g(x)$ is defined to be the function $f(g(x))$, then the “zero vector” is $g(x) = x$. Keep the usual scalar multiplication $cf(x)$ and find two rules that are broken.

Questions 9–18 are about the “subspace requirements”: $x + y$ and cx (and then all linear combinations $cx + dy$) stay in the subspace.

- 9 One requirement can be met while the other fails. Show this by finding
 (a) A set of vectors in $\mathbf{R}^2$ for which $x + y$ stays in the set but $\frac{1}{2}x$ may be outside.
 (b) A set of vectors in $\mathbf{R}^2$ (other than two quarter-planes) for which every cx stays in the set but $x + y$ may be outside.
- 10 Which of the following subsets of $\mathbf{R}^3$ are actually subspaces?
 (a) The plane of vectors (b_1, b_2, b_3) with $b_1 = b_2$.
 (b) The plane of vectors with $b_1 = 1$.
 (c) The vectors with $b_1b_2b_3 = 0$.
 (d) All linear combinations of $v = (1, 4, 0)$ and $w = (2, 2, 2)$.
 (e) All vectors that satisfy $b_1 + b_2 + b_3 = 0$.
 (f) All vectors with $b_1 \leq b_2 \leq b_3$.

- 11 Describe the smallest subspace of the matrix space $\mathbf{M}$ that contains

(a) $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ and $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$ (c) $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ and $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$.

- 12 Let P be the plane in $\mathbf{R}^3$ with equation $x + y - 2z = 4$. The origin $(0, 0, 0)$ is not in P ! Find two vectors in P and check that their sum is not in P .
- 13 Let P_0 be the plane through $(0, 0, 0)$ parallel to the previous plane P . What is the equation for P_0 ? Find two vectors in P_0 and check that their sum is in P_0 .
- 14 The subspaces of $\mathbf{R}^3$ are planes, lines, $\mathbf{R}^3$ itself, or $\mathbf{Z}$ containing only $(0, 0, 0)$.
- Describe the three types of subspaces of $\mathbf{R}^2$.
 - Describe all subspaces of $\mathbf{D}$, the space of 2 by 2 diagonal matrices.
- 15
- The intersection of two planes through $(0, 0, 0)$ is probably a _____ but it could be a _____. It can't be $\mathbf{Z}$!
 - The intersection of a plane through $(0, 0, 0)$ with a line through $(0, 0, 0)$ is probably a _____ but it could be a _____.
 - If S and T are subspaces of $\mathbf{R}^5$, prove that their intersection $S \cap T$ is a subspace of $\mathbf{R}^5$. Here $S \cap T$ consists of the vectors that lie in both subspaces. Check the requirements on $x + y$ and cx .
- 16 Suppose P is a plane through $(0, 0, 0)$ and L is a line through $(0, 0, 0)$. The smallest vector space containing both P and L is either _____ or _____.
- 17
- Show that the set of *invertible* matrices in $\mathbf{M}$ is not a subspace.
 - Show that the set of *singular* matrices in $\mathbf{M}$ is not a subspace.
- 18 True or false (check addition in each case by an example):
- The symmetric matrices in $\mathbf{M}$ (with $A^T = A$) form a subspace.
 - The skew-symmetric matrices in $\mathbf{M}$ (with $A^T = -A$) form a subspace.
 - The unsymmetric matrices in $\mathbf{M}$ (with $A^T \neq A$) form a subspace.

Questions 19–27 are about column spaces $C(A)$ and the equation $Ax = b$.

- 19 Describe the column spaces (lines or planes) of these particular matrices:

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 0 & 0 \end{bmatrix} \quad \text{and} \quad C = \begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 0 & 0 \end{bmatrix}.$$

- 20 For which right sides (find a condition on b_1, b_2, b_3) are these systems solvable?

$$(a) \begin{bmatrix} 1 & 4 & 2 \\ 2 & 8 & 4 \\ -1 & -4 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \quad (b) \begin{bmatrix} 1 & 4 \\ 2 & 9 \\ -1 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}.$$

- 21 Adding row 1 of A to row 2 produces B . Adding column 1 to column 2 produces C . A combination of the columns of (B or C ?) is also a combination of the columns of A . Which two matrices have the same column _____?

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix} \quad \text{and} \quad C = \begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix}.$$

- 22 For which vectors (b_1, b_2, b_3) do these systems have a solution?

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

$$\text{and} \quad \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}.$$

- 23 (Recommended) If we add an extra column $\mathbf{b}$ to a matrix A , then the column space gets larger unless _____. Give an example where the column space gets larger and an example where it doesn't. Why is $A\mathbf{x} = \mathbf{b}$ solvable exactly when the column space *doesn't* get larger—it is the same for A and $[A \ \mathbf{b}]$?
- 24 The columns of AB are combinations of the columns of A . This means: *The column space of AB is contained in (possibly equal to) the column space of A .* Give an example where the column spaces of A and AB are not equal.
- 25 Suppose $A\mathbf{x} = \mathbf{b}$ and $A\mathbf{y} = \mathbf{b}^*$ are both solvable. Then $A\mathbf{z} = \mathbf{b} + \mathbf{b}^*$ is solvable. What is $\mathbf{z}$? This translates into: If $\mathbf{b}$ and $\mathbf{b}^*$ are in the column space $C(A)$, then $\mathbf{b} + \mathbf{b}^*$ is in $C(A)$.
- 26 If A is any 5 by 5 invertible matrix, then its column space is _____. Why?
- 27 True or false (with a counterexample if false):
- (a) The vectors $\mathbf{b}$ that are not in the column space $C(A)$ form a subspace.
 - (b) If $C(A)$ contains only the zero vector, then A is the zero matrix.
 - (c) The column space of $2A$ equals the column space of A .
 - (d) The column space of $A - I$ equals the column space of A (test this).
- 28 Construct a 3 by 3 matrix whose column space contains $(1, 1, 0)$ and $(1, 0, 1)$ but not $(1, 1, 1)$. Construct a 3 by 3 matrix whose column space is only a line.
- 29 If the 9 by 12 system $A\mathbf{x} = \mathbf{b}$ is solvable for every $\mathbf{b}$, then $C(A) =$ _____.

Challenge Problems

- 30** Suppose S and T are two subspaces of a vector space V .
- (a) **Definition:** The **sum** $S + T$ contains all sums $s + t$ of a vector s in S and a vector t in T . Show that $S + T$ satisfies the requirements (addition and scalar multiplication) for a vector space.
 - (b) If S and T are lines in $\mathbf{R}^m$, what is the difference between $S + T$ and $S \cup T$? That union contains all vectors from S or T or both. Explain this statement: *The span of $S \cup T$ is $S + T$.* (Section 3.5 returns to this word “span”.)
- 31** If S is the column space of A and T is $C(B)$, then $S + T$ is the column space of what matrix M ? The columns of A and B and M are all in $\mathbf{R}^m$. (I don't think $A + B$ is always a correct M .)
- 32** Show that the matrices A and $\begin{bmatrix} A & AB \end{bmatrix}$ (with extra columns) have the same column space. But find a square matrix with $C(A^2)$ smaller than $C(A)$. Important point: An n by n matrix has $C(A) = \mathbf{R}^n$ exactly when A is an _____ matrix.

3.2 The Nullspace of A : Solving $Ax = 0$

This section is about the subspace containing all solutions to $Ax = 0$. The m by n matrix A can be square or rectangular. *One immediate solution is $x = 0$.* For invertible matrices this is the only solution. For other matrices, not invertible, there are nonzero solutions to $Ax = 0$. *Each solution x belongs to the nullspace of A .*

Elimination will find all solutions and identify this very important subspace.

The nullspace of A consists of all solutions to $Ax = 0$. These vectors x are in $\mathbb{R}^n$. The nullspace containing all solutions of $Ax = 0$ is denoted by $N(A)$.

Check that the solution vectors form a subspace. Suppose x and y are in the nullspace (this means $Ax = 0$ and $Ay = 0$). The rules of matrix multiplication give $A(x + y) = 0 + 0$. The rules also give $A(cx) = c0$. The right sides are still zero. Therefore $x + y$ and cx are also in the nullspace $N(A)$. Since we can add and multiply without leaving the nullspace, it is a subspace.

To repeat: The solution vectors x have n components. They are vectors in $\mathbb{R}^n$, so *the nullspace is a subspace of $\mathbb{R}^n$* . The column space $C(A)$ is a subspace of $\mathbb{R}^m$.

If the right side b is not zero, the solutions of $Ax = b$ do *not* form a subspace. The vector $x = 0$ is only a solution if $b = 0$. When the set of solutions does not include $x = 0$, it cannot be a subspace. Section 3.4 will show how the solutions to $Ax = b$ (if there are any solutions) are shifted away from the origin by one particular solution.

Example 1 $x + 2y + 3z = 0$ comes from the 1 by 3 matrix $A = [1 \ 2 \ 3]$. This equation $Ax = 0$ produces a plane through the origin $(0, 0, 0)$. The plane is a subspace of $\mathbb{R}^3$. *It is the nullspace of A .*

The solutions to $x + 2y + 3z = 6$ also form a plane, but not a subspace.

Example 2 Describe the nullspace of $A = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}$. This matrix is singular!

Solution Apply elimination to the linear equations $Ax = 0$:

$$\begin{cases} x_1 + 2x_2 = 0 \\ 3x_1 + 6x_2 = 0 \end{cases} \rightarrow \begin{cases} x_1 + 2x_2 = 0 \\ 0 = 0 \end{cases}$$

There is really only one equation. The second equation is the first equation multiplied by 3. In the row picture, the line $x_1 + 2x_2 = 0$ is the same as the line $3x_1 + 6x_2 = 0$. That line is the nullspace $N(A)$. It contains all solutions (x_1, x_2) .

To describe this line of solutions, here is an efficient way. Choose one point on the line (one “*special solution*”). Then all points on the line are multiples of this one. We choose the second component to be $x_2 = 1$ (a special choice). From the equation $x_1 + 2x_2 = 0$, the first component must be $x_1 = -2$. The special solution s is $(-2, 1)$:

**Special
solution**

The nullspace of $A = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}$ contains all multiples of $s = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$.

This is the best way to describe the nullspace, by computing special solutions to $Ax = 0$. This example has one special solution and the nullspace is a line.

The nullspace consists of all combinations of the special solutions.

The plane $x + 2y + 3z = 0$ in Example 1 had *two* special solutions:

$$\begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0 \text{ has the special solutions } s_1 = \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} \text{ and } s_2 = \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix}.$$

Those vectors s_1 and s_2 lie on the plane $x + 2y + 3z = 0$, which is the nullspace of $A = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$. All vectors on the plane are combinations of s_1 and s_2 .

Notice what is special about s_1 and s_2 . They have ones and zeros in the last two components. *Those components are “free” and we choose them specially.* Then the first components -2 and -3 are determined by the equation $Ax = 0$.

The first column of $A = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$ contains the *pivot*, so the first component of x is *not free*. The free components correspond to columns without pivots. This description of special solutions will be completed after one more example.

The special choice (one or zero) is only for the free variables.

Example 3 Describe the nullspaces of these three matrices A, B, C :

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 8 \end{bmatrix} \quad B = \begin{bmatrix} A \\ 2A \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 3 & 8 \\ 2 & 4 \\ 6 & 16 \end{bmatrix} \quad C = \begin{bmatrix} A & 2A \end{bmatrix} = \begin{bmatrix} 1 & 2 & 2 & 4 \\ 3 & 8 & 6 & 16 \end{bmatrix}.$$

Solution The equation $Ax = 0$ has only the zero solution $x = 0$. *The nullspace is $\mathbf{Z}$.* It contains only the single point $x = 0$ in $\mathbf{R}^2$. This comes from elimination:

$$\begin{bmatrix} 1 & 2 \\ 3 & 8 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \text{ yields } \begin{bmatrix} 1 & 2 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \text{ and } \begin{bmatrix} x_1 = 0 \\ x_2 = 0 \end{bmatrix}.$$

A is invertible. There are no special solutions. All columns of this A have pivots.

The rectangular matrix B has the same nullspace $\mathbf{Z}$. The first two equations in $Bx = 0$ again require $x = 0$. The last two equations would also force $x = 0$. When we add extra equations, the nullspace certainly cannot become larger. The extra rows impose more conditions on the vectors x in the nullspace.

The rectangular matrix C is different. It has extra columns instead of extra rows. The solution vector x has *four* components. Elimination will produce pivots in the first two columns of C , but the last two columns are “free”. They don’t have pivots:

$$C = \begin{bmatrix} 1 & 2 & 2 & 4 \\ 3 & 8 & 6 & 16 \end{bmatrix} \text{ becomes } U = \begin{bmatrix} 1 & 2 & 2 & 4 \\ 0 & 2 & 0 & 4 \end{bmatrix}$$

$\begin{matrix} \uparrow & \uparrow & \uparrow & \uparrow \\ \text{pivot columns} & \text{free columns} \end{matrix}$

For the free variables x_3 and x_4 , we make special choices of ones and zeros. First $x_3 = 1$, $x_4 = 0$ and second $x_3 = 0$, $x_4 = 1$. The pivot variables x_1 and x_2 are determined by the

equation $Ux = 0$. We get two special solutions in the nullspace of C (which is also the nullspace of U). The special solutions are s_1 and s_2 :

$$s_1 = \begin{bmatrix} -2 \\ 0 \\ 1 \\ 0 \end{bmatrix} \text{ and } s_2 = \begin{bmatrix} 0 \\ -2 \\ 0 \\ 1 \end{bmatrix} \quad \begin{array}{l} \leftarrow \text{pivot} \\ \leftarrow \text{variables} \\ \leftarrow \text{free} \\ \leftarrow \text{variables} \end{array}$$

One more comment to anticipate what is coming soon. Elimination will not stop at the upper triangular U ! We can continue to make this matrix simpler, in two ways:

1. **Produce zeros above the pivots, by eliminating upward.**
2. **Produce ones in the pivots, by dividing the whole row by its pivot.**

Those steps don't change the zero vector on the right side of the equation. The nullspace stays the same. This nullspace becomes easiest to see when we reach the **reduced row echelon form** R . It has I in the pivot columns:

Reduced form R

$$U = \begin{bmatrix} 1 & 2 & 2 & 4 \\ 0 & 2 & 0 & 4 \end{bmatrix} \text{ becomes } R = \begin{bmatrix} 1 & 0 & 2 & 0 \\ 0 & 1 & 0 & 2 \end{bmatrix}$$

$\uparrow \quad \uparrow$
 now the pivot columns contain I

I subtracted row 2 of U from row 1, and then multiplied row 2 by $\frac{1}{2}$. The original two equations have simplified to $x_1 + 2x_3 = 0$ and $x_2 + 2x_4 = 0$.

The first special solution is still $s_1 = (-2, 0, 1, 0)$, and s_2 is also unchanged. Special solutions are much easier to find from the reduced system $Rx = 0$.

Before moving to m by n matrices A and their nullspaces $N(A)$ and special solutions, allow me to repeat one comment. For many matrices, the only solution to $Ax = 0$ is $x = 0$. Their nullspaces $N(A) = \mathbf{Z}$ contain only that zero vector. The only combination of the columns that produces $b = 0$ is then the "zero combination" or "trivial combination". The solution is trivial (just $x = 0$) but the idea is not trivial.

This case of a zero nullspace $\mathbf{Z}$ is of the greatest importance. It says that the columns of A are **independent**. No combination of columns gives the zero vector (except the zero combination). All columns have pivots, and no columns are free. You will see this idea of independence again ...

Solving $Ax = 0$ by Elimination

This is important. **A is rectangular and we still use elimination.** We solve m equations in n unknowns when $b = 0$. After A is simplified by row operations, we read off the solution (or solutions). Remember the two stages (forward and back) in solving $Ax = 0$:

1. **Forward elimination** takes A to a triangular U (or its reduced form R).
2. **Back substitution** in $Ux = 0$ or $Rx = 0$ produces x .

You will notice a difference in back substitution, when A and U have fewer than n pivots. *We are allowing all matrices in this chapter*, not just the nice ones (which are square matrices with inverses).

Pivots are still nonzero. The columns below the pivots are still zero. But it might happen that a column has no pivot. That free column doesn't stop the calculation. ***Go on to the next column.*** The first example is a 3 by 4 matrix with two pivots:

$$A = \begin{bmatrix} 1 & 1 & 2 & 3 \\ 2 & 2 & 8 & 10 \\ 3 & 3 & 10 & 13 \end{bmatrix}.$$

Certainly $a_{11} = 1$ is the first pivot. Clear out the 2 and 3 below that pivot:

$$A \rightarrow \begin{bmatrix} 1 & 1 & 2 & 3 \\ 0 & 0 & 4 & 4 \\ 0 & 0 & 4 & 4 \end{bmatrix} \quad \begin{array}{l} \text{(subtract } 2 \times \text{ row 1)} \\ \text{(subtract } 3 \times \text{ row 1)} \end{array}$$

The second column has a zero in the pivot position. We look below the zero for a nonzero entry, ready to do a row exchange. *The entry below that position is also zero.* Elimination can do nothing with the second column. This signals trouble, which we expect anyway for a rectangular matrix. There is no reason to quit, and we go on to the third column.

The second pivot is 4 (but it is in the third column). Subtracting row 2 from row 3 clears out that column below the pivot. **The pivot columns are 1 and 3:**

$$\text{Triangular } U : U = \begin{bmatrix} 1 & 1 & 2 & 3 \\ 0 & 0 & 4 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \begin{array}{l} \text{Only two pivots} \\ \text{The last equation} \\ \text{became } 0 = 0 \end{array}$$

The fourth column also has a zero in the pivot position—but nothing can be done. There is no row below it to exchange, and forward elimination is complete. The matrix has three rows, four columns, and *only two pivots*. The original $Ax = 0$ seemed to involve three different equations, but the third equation is the sum of the first two. It is automatically satisfied ($0 = 0$) when the first two equations are satisfied. Elimination reveals the inner truth about a system of equations. Soon we push on from U to R .

Now comes back substitution, to find all solutions to $Ux = 0$. With four unknowns and only two pivots, there are many solutions. The question is how to write them all down. A good method is to separate the ***pivot variables*** from the ***free variables***.

P	The <i>pivot</i> variables are x_1 and x_3 .	Columns 1 and 3 contain pivots.
F	The <i>free</i> variables are x_2 and x_4 .	Columns 2 and 4 have no pivots.

The free variables x_2 and x_4 can be given any values whatsoever. Then back substitution finds the pivot variables x_1 and x_3 . (In Chapter 2 no variables were free. When A is invertible, all variables are pivot variables.) The simplest choices for the free variables are ones and zeros. Those choices give the *special solutions*.

Special solutions to $x_1 + x_2 + 2x_3 + 3x_4 = 0$ and $4x_3 + 4x_4 = 0$

- Set $x_2 = 1$ and $x_4 = 0$. By back substitution $x_3 = 0$. Then $x_1 = -1$.
- Set $x_2 = 0$ and $x_4 = 1$. By back substitution $x_3 = -1$. Then $x_1 = -1$.

These special solutions solve $Ux = 0$ and therefore $Ax = 0$. They are in the nullspace. The good thing is that *every solution is a combination of the special solutions*.

$$\begin{array}{l} \text{Complete solution} \\ \text{to } Ax = 0 \end{array} \quad x = x_2 \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -1 \\ 0 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} -x_2 - x_4 \\ x_2 \\ -x_4 \\ x_4 \end{bmatrix}. \quad (1)$$

special special complete

Please look again at that answer. It is the main goal of this section. The vector $s_1 = (-1, 1, 0, 0)$ is the special solution when $x_2 = 1$ and $x_4 = 0$. The second special solution has $x_2 = 0$ and $x_4 = 1$. **All solutions are linear combinations of s_1 and s_2 .** The special solutions are in the nullspace $N(A)$, and their combinations fill out the whole nullspace.

The MATLAB code **nullbasis** computes these special solutions. They go into the columns of a **nullspace matrix** N . The complete solution to $Ax = 0$ is a combination of those columns. Once we have the special solutions, we have the whole nullspace.

There is a special solution for each free variable. If no variables are free—this means there are n pivots—then the only solution to $Ux = 0$ and $Ax = 0$ is the trivial solution $x = 0$. All variables are pivot variables. In that case the nullspaces of A and U contain only the zero vector. With no free variables, and pivots in every column, the output from **nullbasis** is an empty matrix. The nullspace with n pivots is $\{0\}$.

Example 4 Find the nullspace of $U = \begin{bmatrix} 1 & 5 & 7 \\ 0 & 0 & 9 \end{bmatrix}$.

The second column of U has no pivot. So x_2 is free. The special solution has $x_2 = 1$. Back substitution into $9x_3 = 0$ gives $x_3 = 0$. Then $x_1 + 5x_2 = 0$ or $x_1 = -5$. The solutions to $Ux = 0$ are multiples of one special solution:

$$x = x_2 \begin{bmatrix} -5 \\ 1 \\ 0 \end{bmatrix}$$

The nullspace of U is a line in $\mathbb{R}^3$.
It contains multiples of the special solution $s = (-5, 1, 0)$.
One variable is free, and $N = \text{nullbasis}(U)$ has one column s .

In a minute elimination will get zeros above the pivots and ones in the pivots. By continuing elimination on U , the 7 is removed and the pivot changes from 9 to 1.

The final result will be the **reduced row echelon form R** :

$$U = \begin{bmatrix} 1 & 5 & 7 \\ 0 & 0 & 9 \end{bmatrix} \text{ reduces to } R = \begin{bmatrix} 1 & 5 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \text{rref}(U).$$

This makes it even clearer that the special solution (column of N) is $s = (-5, 1, 0)$.

Echelon Matrices

Forward elimination goes from A to U . It acts by row operations, including row exchanges. It goes on to the next column when no pivot is available in the current column. The m by n “staircase” U is an **echelon matrix**.

Here is a 4 by 7 echelon matrix with the three pivots p highlighted in boldface:

$$U = \begin{bmatrix} \mathbf{p} & x & x & x & x & x & x \\ 0 & \mathbf{p} & x & x & x & x & x \\ 0 & 0 & 0 & 0 & 0 & \mathbf{p} & x \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Three pivot variables x_1, x_2, x_6
Four free variables x_3, x_4, x_5, x_7
Four special solutions in $N(U)$

Question What are the column space and the nullspace for this matrix?

Answer The columns have four components so they lie in $\mathbf{R}^4$. (Not in $\mathbf{R}^3$!) The fourth component of every column is zero. Every combination of the columns—every vector in the column space—has fourth component zero. *The column space $C(U)$ consists of all vectors of the form $(b_1, b_2, b_3, 0)$.* For those vectors we can solve $Ux = b$ by back substitution. These vectors b are all possible combinations of the seven columns.

The nullspace $N(U)$ is a subspace of $\mathbf{R}^7$. The solutions to $Ux = 0$ are all the combinations of the four special solutions—one for each free variable:

1. Columns 3, 4, 5, 7 have no pivots. So the free variables are x_3, x_4, x_5, x_7 .
2. Set one free variable to 1 and set the other free variables to zero.
3. Solve $Ux = 0$ for the pivot variables x_1, x_2, x_6 .
4. This gives one of the four special solutions in the nullspace matrix N .

The nonzero rows of an echelon matrix go down in a staircase pattern. The pivots are the first nonzero entries in those rows. There is a column of zeros below every pivot.

Counting the pivots leads to an extremely important theorem. Suppose A has more columns than rows. **With $n > m$ there is at least one free variable.** The system $Ax = 0$ has at least one special solution. This solution is *not zero*!

Suppose $Ax = 0$ has more unknowns than equations ($n > m$, more columns than rows). Then there are **nonzero solutions**. There must be free columns, without pivots.

A short wide matrix ($n > m$) always has nonzero vectors in its nullspace. There must be at least $n - m$ free variables, since the number of pivots cannot exceed m . (The matrix only has m rows, and a row never has two pivots.) Of course a row might have *no* pivot—which means an extra free variable. But here is the point: When there is a free variable, it can be set to 1. Then the equation $Ax = 0$ has a nonzero solution.

To repeat: There are at most m pivots. With $n > m$, the system $Ax = 0$ has a nonzero solution. Actually there are infinitely many solutions, since any multiple cx is also a solution. The nullspace contains at least a line of solutions. With two free variables, there are two special solutions and the nullspace is even larger.

The nullspace is a subspace. Its “dimension” is the number of free variables. This central idea—the **dimension** of a subspace—is defined and explained in this chapter.

The Reduced Row Echelon Matrix R

From an echelon matrix U we go one more step. Continue with a 3 by 4 example:

$$U = \begin{bmatrix} 1 & 1 & 2 & 3 \\ 0 & 0 & 4 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

We can divide the second row by 4. Then both pivots equal 1. We can subtract 2 times this new row $[0 \ 0 \ 1 \ 1]$ from the row above. The reduced row echelon matrix R has zeros above the pivots as well as below:

Reduced row
echelon matrix

$$R = \text{rref}(A) = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Pivot rows
contain I

R has 1's as pivots. Zeros above pivots come from upward elimination.

Important If A is invertible, its reduced row echelon form is the identity matrix $R = I$. This is the ultimate in row reduction. Of course the nullspace is then $\mathbf{Z}$.

The zeros in R make it easy to find the special solutions (the same as before):

1. Set $x_2 = 1$ and $x_4 = 0$. Solve $Rx = 0$. Then $x_1 = -1$ and $x_3 = 0$.

Those numbers -1 and 0 are sitting in column 2 of R (with plus signs).

2. Set $x_2 = 0$ and $x_4 = 1$. Solve $Rx = 0$. Then $x_1 = -1$ and $x_3 = -1$.

Those numbers -1 and -1 are sitting in column 4 (with plus signs).

By reversing signs we can read off the special solutions directly from R . The nullspace $N(A) = N(U) = N(R)$ contains all combinations of the special solutions:

$$x = x_2 \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -1 \\ 0 \\ -1 \\ 1 \end{bmatrix} = (\text{complete solution of } Ax = 0).$$

The next section of the book moves firmly from U to the row reduced form R . The MATLAB command `[ R, pivcol ] = rref(A)` produces R and also a list of the pivot columns.

■ REVIEW OF THE KEY IDEAS ■

1. The nullspace $N(A)$ is a subspace of $\mathbb{R}^n$. It contains all solutions to $Ax = 0$.
2. Elimination produces an echelon matrix U , and then a row reduced R , with pivot columns and free columns.
3. Every free column of U or R leads to a special solution. The free variable equals 1 and the other free variables equal 0. Back substitution solves $Ax = 0$.
4. The complete solution to $Ax = 0$ is a combination of the special solutions.
5. If $n > m$ then A has at least one column without pivots, giving a special solution. So there are nonzero vectors x in the nullspace of this rectangular A .

■ WORKED EXAMPLES ■

3.2 A Create a 3 by 4 matrix whose special solutions to $Ax = 0$ are s_1 and s_2 :

$$s_1 = \begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \end{bmatrix} \quad \text{and} \quad s_2 = \begin{bmatrix} -2 \\ 0 \\ -6 \\ 1 \end{bmatrix} \quad \begin{array}{l} \text{pivot columns 1 and 3} \\ \text{free variables } x_2 \text{ and } x_4 \end{array}$$

You could create the matrix A in row reduced form R . Then describe all possible matrices A with the required nullspace $N(A) = \text{all combinations of } s_1 \text{ and } s_2$.

Solution The reduced matrix R has pivots = 1 in columns 1 and 3. There is no third pivot, so the third row of R is all zeros. The free columns 2 and 4 will be combinations of the pivot columns:

$$R = \begin{bmatrix} 1 & 3 & 0 & 2 \\ 0 & 0 & 1 & 6 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \text{has} \quad Rs_1 = 0 \quad \text{and} \quad Rs_2 = 0.$$

The entries 3, 2, 6 in R are the negatives of $-3, -2, -6$ in the special solutions!

R is only one matrix (one possible A) with the required nullspace. We could do any elementary operations on R —exchange rows, multiply a row by any $c \neq 0$, subtract any multiple of one row from another. **R can be multiplied (on the left) by any invertible matrix, without changing its nullspace.**

Every 3 by 4 matrix has at least one special solution. *These matrices have two.*

3.2 B Find the special solutions and describe the *complete solution* to $Ax = \mathbf{0}$ for

$$A_1 = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad A_2 = \begin{bmatrix} 3 & 6 \\ 1 & 2 \end{bmatrix} \quad A_3 = [A_2 \quad A_2]$$

Which are the pivot columns? Which are the free variables? What is R in each case?

Solution $A_1x = \mathbf{0}$ has four special solutions. They are the columns s_1, s_2, s_3, s_4 of the 4 by 4 identity matrix. The nullspace is all of $\mathbf{R}^4$. The complete solution to $A_1x = \mathbf{0}$ is any $x = c_1s_1 + c_2s_2 + c_3s_3 + c_4s_4$ in $\mathbf{R}^4$. There are no pivot columns; all variables are free; the reduced R is the same zero matrix as A_1 .

$A_2x = \mathbf{0}$ has only one special solution $s = (-2, 1)$. The multiples $x = cs$ give the complete solution. The first column of A_2 is its pivot column, and x_2 is the free variable. The row reduced matrices R_2 for A_2 and R_3 for $A_3 = [A_2 \quad A_2]$ have 1's in the pivot:

$$A_2 = \begin{bmatrix} 3 & 6 \\ 1 & 2 \end{bmatrix} \rightarrow R_2 = \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix} \quad [A_2 \quad A_2] \rightarrow R_3 = \begin{bmatrix} 1 & 2 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Notice that R_3 has only one pivot column (the first column). All the variables x_2, x_3, x_4 are free. There are three special solutions to $A_3x = \mathbf{0}$ (and also $R_3x = \mathbf{0}$):

$s_1 = (-2, 1, 0, 0)$ $s_2 = (-1, 0, 1, 0)$ $s_3 = (-2, 0, 0, 1)$ **Complete** $x = c_1s_1 + c_2s_2 + c_3s_3$.

With r pivots, A has $n - r$ free variables. $Ax = \mathbf{0}$ has $n - r$ special solutions.

Problem Set 3.2

Questions 1–4 and 5–8 are about the matrices in Problems 1 and 5.

1 Reduce these matrices to their ordinary echelon forms U :

$$(a) \quad A = \begin{bmatrix} 1 & 2 & 2 & 4 & 6 \\ 1 & 2 & 3 & 6 & 9 \\ 0 & 0 & 1 & 2 & 3 \end{bmatrix} \quad (b) \quad B = \begin{bmatrix} 2 & 4 & 2 \\ 0 & 4 & 4 \\ 0 & 8 & 8 \end{bmatrix}.$$

Which are the free variables and which are the pivot variables?

- 2 For the matrices in Problem 1, find a special solution for each free variable. (Set the free variable to 1. Set the other free variables to zero.)
- 3 By combining the special solutions in Problem 2, describe every solution to $Ax = \mathbf{0}$ and $Bx = \mathbf{0}$. The nullspace contains only $x = \mathbf{0}$ when there are no _____.
- 4 By further row operations on each U in Problem 1, find the reduced echelon form R . *True or false:* The nullspace of R equals the nullspace of U .
- 5 By row operations reduce each matrix to its echelon form U . Write down a 2 by 2 lower triangular L such that $B = LU$.

$$(a) A = \begin{bmatrix} -1 & 3 & 5 \\ -2 & 6 & 10 \end{bmatrix} \quad (b) B = \begin{bmatrix} -1 & 3 & 5 \\ -2 & 6 & 7 \end{bmatrix}.$$

- 6 For the same A and B , find the special solutions to $Ax = 0$ and $Bx = 0$. For an m by n matrix, the number of pivot variables plus the number of free variables is ____.
- 7 In Problem 5, describe the nullspaces of A and B in two ways. Give the equations for the plane or the line, and give all vectors x that satisfy those equations as combinations of the special solutions.
- 8 Reduce the echelon forms U in Problem 5 to R . For each R draw a box around the identity matrix that is in the pivot rows and pivot columns.

Questions 9–17 are about free variables and pivot variables.

- 9 True or false (with reason if true or example to show it is false):
- (a) A square matrix has no free variables.
 - (b) An invertible matrix has no free variables.
 - (c) An m by n matrix has no more than n pivot variables.
 - (d) An m by n matrix has no more than m pivot variables.
- 10 Construct 3 by 3 matrices A to satisfy these requirements (if possible):
- (a) A has no zero entries but $U = I$.
 - (b) A has no zero entries but $R = I$.
 - (c) A has no zero entries but $R = U$.
 - (d) $A = U = 2R$.
- 11 Put as many 1's as possible in a 4 by 7 echelon matrix U whose pivot columns are
- (a) 2, 4, 5
 - (b) 1, 3, 6, 7
 - (c) 4 and 6.
- 12 Put as many 1's as possible in a 4 by 8 *reduced* echelon matrix R so that the free columns are
- (a) 2, 4, 5, 6
 - (b) 1, 3, 6, 7, 8.
- 13 Suppose column 4 of a 3 by 5 matrix is all zero. Then x_4 is certainly a ____ variable. The special solution for this variable is the vector $x =$ ____.
- 14 Suppose the first and last columns of a 3 by 5 matrix are the same (not zero). Then ____ is a free variable. Find the special solution for this variable.

- 15 Suppose an m by n matrix has r pivots. The number of special solutions is _____. The nullspace contains only $\mathbf{x} = \mathbf{0}$ when $r =$ _____. The column space is all of $\mathbf{R}^m$ when $r =$ _____.
- 16 The nullspace of a 5 by 5 matrix contains only $\mathbf{x} = \mathbf{0}$ when the matrix has _____ pivots. The column space is $\mathbf{R}^5$ when there are _____ pivots. Explain why.
- 17 The equation $x - 3y - z = 0$ determines a plane in $\mathbf{R}^3$. What is the matrix A in this equation? Which are the free variables? The special solutions are $(3, 1, 0)$ and _____.
- 18 (Recommended) The plane $x - 3y - z = 12$ is parallel to the plane $x - 3y - z = 0$ in Problem 17. One particular point on this plane is $(12, 0, 0)$. All points on the plane have the form (fill in the first components)

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + y \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + z \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

- 19 Prove that U and $A = LU$ have the same nullspace when L is invertible:

If $U\mathbf{x} = \mathbf{0}$ then $LU\mathbf{x} = \mathbf{0}$. If $LU\mathbf{x} = \mathbf{0}$, how do you know $U\mathbf{x} = \mathbf{0}$?

- 20 Suppose column 1 + column 3 + column 5 = $\mathbf{0}$ in a 4 by 5 matrix with four pivots. Which column is sure to have no pivot (and which variable is free)? What is the special solution? What is the nullspace?

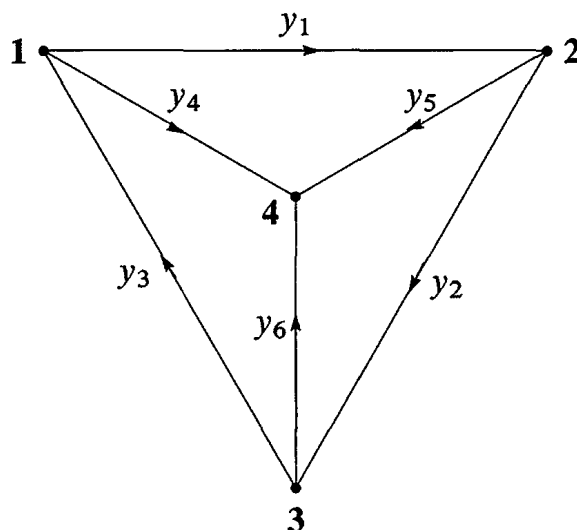
Questions 21–28 ask for matrices (if possible) with specific properties.

- 21 Construct a matrix whose nullspace consists of all combinations of $(2, 2, 1, 0)$ and $(3, 1, 0, 1)$.
- 22 Construct a matrix whose nullspace consists of all multiples of $(4, 3, 2, 1)$.
- 23 Construct a matrix whose column space contains $(1, 1, 5)$ and $(0, 3, 1)$ and whose nullspace contains $(1, 1, 2)$.
- 24 Construct a matrix whose column space contains $(1, 1, 0)$ and $(0, 1, 1)$ and whose nullspace contains $(1, 0, 1)$ and $(0, 0, 1)$.
- 25 Construct a matrix whose column space contains $(1, 1, 1)$ and whose nullspace is the line of multiples of $(1, 1, 1, 1)$.
- 26 Construct a 2 by 2 matrix whose nullspace equals its column space. This is possible.
- 27 Why does no 3 by 3 matrix have a nullspace that equals its column space?
- 28 If $AB = 0$ then the column space of B is contained in the _____ of A . Give an example of A and B .

- 29** The reduced form R of a 3 by 3 matrix with randomly chosen entries is almost sure to be _____. What R is virtually certain if the random A is 4 by 3?
- 30** Show by example that these three statements are generally *false*:
- (a) A and A^T have the same nullspace.
 - (b) A and A^T have the same free variables.
 - (c) If R is the reduced form $\text{rref}(A)$ then R^T is $\text{rref}(A^T)$.
- 31** If the nullspace of A consists of all multiples of $x = (2, 1, 0, 1)$, how many pivots appear in U ? What is R ?
- 32** If the special solutions to $Rx = 0$ are in the columns of these N , go backward to find the nonzero rows of the reduced matrices R :
- $$N = \begin{bmatrix} 2 & 3 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \text{and} \quad N = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \quad \text{and} \quad N = \begin{bmatrix} \\ \\ \end{bmatrix} \quad (\text{empty } 3 \text{ by } 1).$$
- 33** (a) What are the five 2 by 2 reduced echelon matrices R whose entries are all 0's and 1's?
- (b) What are the eight 1 by 3 matrices containing only 0's and 1's? Are all eight of them reduced echelon matrices R ?
- 34** Explain why A and $-A$ always have the same reduced echelon form R .

Challenge Problems

- 35** If A is 4 by 4 and invertible, describe all vectors in the nullspace of the 4 by 8 matrix $B = [A \ A]$.
- 36** How is the nullspace $N(C)$ related to the spaces $N(A)$ and $N(B)$, if $C = \begin{bmatrix} A \\ B \end{bmatrix}$?
- 37** Kirchhoff's Law says that *current in* = *current out* at every node. This network has six currents $y_1, \dots, y_6$ (the arrows show the positive direction, each y_i could be positive or negative). Find the four equations $Ay = 0$ for Kirchhoff's Law at the four nodes. Find three special solutions in the nullspace of A .



3.3 The Rank and the Row Reduced Form

The numbers m and n give the size of a matrix—but not necessarily the *true size* of a linear system. An equation like $0 = 0$ should not count. If there are two identical rows in A , the second one disappears in elimination. Also if row 3 is a combination of rows 1 and 2, then row 3 will become all zeros in the triangular U and the reduced echelon form R . We don't want to count rows of zeros. **The true size of A is given by its rank:**

DEFINITION *The rank of A is the number of pivots. This number is r .*

That definition is computational, and I would like to say more about the rank r . The matrix will eventually be reduced to r nonzero rows. Start with a 3 by 4 example.

Four columns

How many pivots?

$$A = \begin{bmatrix} 1 & 1 & 2 & 4 \\ 1 & 2 & 2 & 5 \\ 1 & 3 & 2 & 6 \end{bmatrix}. \quad (1)$$

The first two columns are $(1, 1, 1)$ and $(1, 2, 3)$, going in different directions. Those will be pivot columns. The third column $(2, 2, 2)$ is a multiple of the first. We won't see a pivot in that third column. The fourth column $(4, 5, 6)$ is a combination of the first three (their sum). That column will also be without a pivot.

The fourth column is actually a combination $3(1, 1, 1) + (1, 2, 3)$ of the two pivot columns. **Every “free column” is a combination of earlier pivot columns.** It is the **special solutions s** that tell us those combinations of pivot columns:

$$\begin{array}{ll} \text{Column 3} = 2 (\text{column 1}) & s_1 = (-2, 0, 1, 0) \quad As_1 = 0 \\ \text{Column 4} = 3 (\text{column 1}) + 1 (\text{column 2}) & s_2 = (-3, -1, 0, 1) \quad As_2 = 0 \end{array}$$

With nice numbers we can see the right combinations. The systematic way to find s is by elimination! This will change the columns but it won't change the combinations, because $Ax = 0$ is equivalent to $Ux = 0$ and also $Rx = 0$. I will go from A to U and then to R :

$$\begin{bmatrix} 1 & 1 & 2 & 4 \\ 1 & 2 & 2 & 5 \\ 1 & 3 & 2 & 6 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 2 & 4 \\ 0 & 1 & 0 & 1 \\ 0 & 2 & 0 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 2 & 4 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} = U$$

U already shows the two pivots in the pivot columns. **The rank of A (and U) is 2.** Continuing to R we see the combinations of pivot columns that produce the free columns:

$$U = \begin{bmatrix} 1 & 1 & 2 & 4 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow[\text{row 1} - \text{row 2}]{\text{Subtract}} R = \begin{bmatrix} 1 & 0 & 2 & 3 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad (2)$$

Clearly the $(3, 1, 0)$ column equals $3 (\text{column 1}) + (\text{column 2})$. Moving all columns to the “left side” will reverse signs to -3 and -1 , which go in the special solution s :

$$-3 (\text{column 1}) - (\text{column 2}) + (\text{column 4}) = 0 \quad s = (-3, -1, 0, 1).$$

Rank One

Matrices of **rank one** have only *one pivot*. When elimination produces zero in the first column, it produces zero in all the columns. *Every row is a multiple of the pivot row*. At the same time, every column is a multiple of the pivot column!

$$\text{Rank one matrix} \quad A = \begin{bmatrix} 1 & 3 & 10 \\ 2 & 6 & 20 \\ 3 & 9 & 30 \end{bmatrix} \longrightarrow R = \begin{bmatrix} 1 & 3 & 10 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

The column space of a rank one matrix is “one-dimensional”. Here all columns are on the line through $\mathbf{u} = (1, 2, 3)$. The columns of A are $\mathbf{u}$ and $3\mathbf{u}$ and $10\mathbf{u}$. Put those numbers into the row $\mathbf{v}^T = [1 \ 3 \ 10]$ and you have the special rank one form $A = \mathbf{u}\mathbf{v}^T$:

$$A = \text{column times row} = \mathbf{u}\mathbf{v}^T \quad \begin{bmatrix} 1 & 3 & 10 \\ 2 & 6 & 20 \\ 3 & 9 & 30 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \begin{bmatrix} 1 & 3 & 10 \end{bmatrix} \quad (3)$$

With rank one, the solutions to $A\mathbf{x} = \mathbf{0}$ are easy to understand. That equation $\mathbf{u}(\mathbf{v}^T\mathbf{x}) = \mathbf{0}$ leads us to $\mathbf{v}^T\mathbf{x} = 0$. All vectors $\mathbf{x}$ in the nullspace must be orthogonal to $\mathbf{v}$ in the row space. This is the geometry: *row space = line, nullspace = perpendicular plane*. Now describe the special solutions with numbers:

$$\begin{array}{ll} \text{Pivot row } [1 \ 3 \ 10] & \\ \text{Pivot variable } x_1 & s_1 = \begin{bmatrix} -3 \\ 1 \\ 0 \end{bmatrix} \quad s_2 = \begin{bmatrix} -10 \\ 0 \\ 1 \end{bmatrix} \\ \text{Free variables } x_2 \text{ and } x_3 & \end{array}$$

The nullspace contains all combinations of s_1 and s_2 . This produces the plane $x + 3y + 10z = 0$, perpendicular to the row $(1, 3, 10)$. **Nullspace (plane) perpendicular to row space (line).**

Example 1 When all rows are multiples of one pivot row, the rank is $r = 1$:

$$\begin{bmatrix} 1 & 3 & 4 \\ 2 & 6 & 8 \end{bmatrix} \text{ and } \begin{bmatrix} 0 & 3 \\ 0 & 5 \end{bmatrix} \text{ and } \begin{bmatrix} 5 \\ 2 \end{bmatrix} \text{ and } [6] \text{ all have rank 1.}$$

For those matrices, the reduced row echelon $R = \text{rref}(A)$ can be checked by eye:

$$R = \begin{bmatrix} 1 & 3 & 4 \\ 0 & 0 & 0 \end{bmatrix} \text{ and } \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \text{ and } \begin{bmatrix} 1 \\ 0 \end{bmatrix} \text{ and } [1] \text{ have only one pivot.}$$

Our second definition of rank will be at a higher level. It deals with entire rows and entire columns—vectors and not just numbers. The matrices A and U and R have r *independent* rows (the pivot rows). They also have r *independent* columns (the pivot columns). Section 3.5 says what it means for rows or columns to be independent.

A third definition of rank, at the top level of linear algebra, will deal with *spaces* of vectors. **The rank r is the “dimension” of the column space.** It is also the dimension of the row space. The great thing is that r also reveals the dimension of the nullspace.

The Pivot Columns

The pivot columns of R have 1's in the pivots and 0's everywhere else. The r pivot columns taken together contain an r by r identity matrix I . It sits above $m - r$ rows of zeros. The numbers of the pivot columns are in the list *pivcol*.

The pivot columns of A are probably not obvious from A itself. But their column numbers are given by the *same list pivcol*. The r columns of A that eventually have pivots (in U and R) are the pivot columns of A . This example has $\text{pivcol} = (1, 3)$:

$$\begin{array}{l} \text{Pivot} \\ \text{Columns} \end{array} \quad A = \begin{bmatrix} 1 & 3 & 0 & 2 & -1 \\ 0 & 0 & 1 & 4 & -3 \\ 1 & 3 & 1 & 6 & -4 \end{bmatrix} \text{ yields } R = \begin{bmatrix} 1 & 3 & 0 & 2 & -1 \\ 0 & 0 & 1 & 4 & -3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

The column spaces of A and R are different! All columns of this R end with zeros. Elimination subtracts rows 1 and 2 of A from row 3, to produce that zero row in R :

$$\begin{array}{l} EA = R \\ A = E^{-1}R \end{array} \quad E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & -1 & 1 \end{bmatrix} \quad \text{and} \quad E^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}.$$

The r pivot columns of A are also the first r columns of E^{-1} . The r by r identity matrix inside R just picks out the first r columns of E^{-1} as columns of $A = E^{-1}R$.

One more fact about pivot columns. Their definition has been purely computational, based on R . Here is a direct mathematical description of the pivot columns of A :

The pivot columns are not combinations of earlier columns. The free columns *are* combinations of earlier columns. These combinations are the special solutions!

A pivot column of R (with 1 in the pivot row) cannot be a combination of earlier columns (with 0's in that row). The same column of A can't be a combination of earlier columns, because $Ax = 0$ exactly when $Rx = 0$.

Now we look at the special solution x from each free column.

The Special Solutions

Each special solution to $Ax = 0$ and $Rx = 0$ has one free variable equal to 1. The other free variables in x are all zero. The solutions come directly from the echelon form R :

$$\begin{array}{l} \text{Free columns} \\ \text{Free variables} \\ \text{in boldface} \end{array} \quad Rx = \begin{bmatrix} 1 & 3 & 0 & 2 & -1 \\ 0 & 0 & 1 & 4 & -3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Set the first free variable to $x_2 = 1$ with $x_4 = x_5 = 0$. The equations give the pivot variables $x_1 = -3$ and $x_3 = 0$. The special solution is $s_1 = (-3, 1, 0, 0, 0)$.

The next special solution has $x_4 = 1$. The other free variables are $x_2 = x_5 = 0$. The solution is $s_2 = (-2, 0, -4, 1, 0)$. Notice -2 and -4 in R , with plus signs.

The third special solution has $x_5 = 1$. With $x_2 = 0$ and $x_4 = 0$ we find $s_3 = (1, 0, 3, 0, 1)$. The numbers $x_1 = 1$ and $x_3 = 3$ are in column 5 of R , again with opposite signs. This is a general rule as we soon verify. The nullspace matrix N contains the three special solutions in its columns, so $AN = \text{zero matrix}$:

Nullspace matrix
 $n - r = 5 - 2$
3 special solutions

$$N = \begin{bmatrix} -3 & -2 & 1 \\ 1 & 0 & 0 \\ 0 & -4 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{array}{l} \text{not free} \\ \text{free} \\ \text{not free} \\ \text{free} \\ \text{free} \end{array}$$

The linear combinations of these three columns give all vectors in the nullspace. This is the complete solution to $Ax = 0$ (and $Rx = 0$). Where R had the identity matrix (2 by 2) in its pivot columns, N has the identity matrix (3 by 3) in its free rows.

There is a special solution for every free variable. Since r columns have pivots, that leaves $n - r$ free variables. This is the key to $Ax = 0$ and the nullspace:

$Ax = 0$ has r pivots and $n - r$ free variables: n columns minus r pivot columns. The **nullspace matrix** N contains the $n - r$ special solutions. Then $AN = 0$.

When we introduce the idea of “independent” vectors, we will show that the special solutions are independent. You can see in N that no column is a combination of the other columns. The beautiful thing is that the count is exactly right:

$Ax = 0$ has r independent equations so it has $n - r$ independent solutions.

The special solutions are easy for $Rx = 0$. Suppose that the first r columns are the pivot columns. Then the reduced row echelon form looks like

$$R = \begin{bmatrix} I & F \\ 0 & 0 \end{bmatrix} \begin{array}{l} r \text{ pivot rows} \\ m - r \text{ zero rows} \end{array} \quad (4)$$

r pivot columns $n - r$ free columns

The pivot variables in the $n - r$ special solutions come by changing F to $-F$:

$$\text{Nullspace matrix } N = \begin{bmatrix} -F \\ I \end{bmatrix} \begin{array}{l} r \text{ pivot variables} \\ n - r \text{ free variables} \end{array} \quad (5)$$

Check $RN = 0$. The first block row of RN is $(I \text{ times } -F) + (F \text{ times } I) = \text{zero}$. The columns of N solve $Rx = 0$. When the free part of $Rx = 0$ moves to the right side,

the left side just holds the identity matrix:

$$Rx = 0 \quad \text{means} \quad I \begin{bmatrix} \text{pivot} \\ \text{variables} \end{bmatrix} = -F \begin{bmatrix} \text{free} \\ \text{variables} \end{bmatrix}. \quad (6)$$

In each special solution, the free variables are a column of I . Then the pivot variables are a column of $-F$. Those special solutions give the nullspace matrix N .

The idea is still true if the pivot columns are mixed in with the free columns. Then I and F are mixed together. You can still see $-F$ in the solutions. Here is an example where $I = [1]$ comes first and $F = [2 \ 3]$ comes last.

Example 2 The special solutions of $Rx = x_1 + 2x_2 + 3x_3 = 0$ are the columns of N :

$$R = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \quad N = \begin{bmatrix} -F \\ I \end{bmatrix} = \begin{bmatrix} -2 & -3 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

The rank is one. There are $n - r = 3 - 1$ special solutions $(-2, 1, 0)$ and $(-3, 0, 1)$.

Final Note How can I write confidently about R not knowing which steps MATLAB will take? A could be reduced to R in different ways. Very likely you and Mathematica and Maple would do the elimination differently. The key is that **the final R is always the same**. The original A completely determines the I and F and zero rows in R .

For proof I will determine the pivot columns (which locate I) and free columns (which contain F) in an “algebra way”—two rules that have nothing to do with any particular elimination steps. Here are those rules:

1. The pivot columns *are not* combinations of earlier columns of A .
2. The free columns *are* combinations of earlier columns (F tells the combinations).

A small example with rank one will show two E ’s that produce the correct $EA = R$:

$$A = \begin{bmatrix} 2 & 2 \\ 1 & 1 \end{bmatrix} \quad \text{reduces to} \quad R = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} = \text{rref}(A) \quad \text{and no other } R.$$

You could multiply row 1 of A by $\frac{1}{2}$, and subtract row 1 from row 2:

$$\text{Two steps give } E \quad \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1/2 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1/2 & 0 \\ -1/2 & 1 \end{bmatrix} = E.$$

Or you could exchange rows in A , and then subtract 2 times row 1 from row 2:

$$\text{Two different steps give } E_{\text{new}} \quad \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & -2 \end{bmatrix} = E_{\text{new}}.$$

Multiplication gives $EA = R$ and also $E_{\text{new}}A = R$. **Different E ’s but the same R .**

Codes for Row Reduction

There is no way that `rref` will ever come close in importance to `lu`. The Teaching Code `elim` for this book uses `rref`. Of course `rref(R)` would give R again!

MATLAB: $[R, pivcol] = rref(A)$ Teaching Code: $[E, R] = elim(A)$

The extra output *pivcol* gives the numbers of the pivot columns. They are the same in A and R . The extra output E in the Teaching Code is an m by m **elimination matrix** that puts the original A (whatever it was) into its row reduced form R :

$$E A = R.$$

The square matrix E is the product of elementary matrices E_{ij} and also P_{ij} and D^{-1} . P_{ij} exchanges rows. The diagonal D^{-1} divides rows by their pivots to produce 1's.

If we want E , we can apply row reduction to the matrix $[A \ I]$ with $n + m$ columns. All the elementary matrices that multiply A (to produce R) will also multiply I (to produce E). The whole augmented matrix is being multiplied by E :

$$E [A \ I] = [R \ E] \quad (7)$$

This is exactly what “Gauss-Jordan” did in Chapter 2 to compute A^{-1} . **When A is square and invertible, its reduced row echelon form is I .** Then $EA = R$ becomes $EA = I$. In this invertible case, E is A^{-1} . This chapter is going further, to every A .

■ REVIEW OF THE KEY IDEAS ■

1. The rank r of A is the number of pivots (which are 1's in $R = rref(A)$).
2. The r pivot columns of A and R are in the same list *pivcol*.
3. Those r pivot columns are not combinations of earlier columns.
4. The $n - r$ free columns *are* combinations of earlier columns (pivot columns).
5. Those combinations (using $-F$ taken from R) give the $n - r$ special solutions to $Ax = 0$ and $Rx = 0$. They are the $n - r$ columns of the nullspace matrix N .

■ WORKED EXAMPLES ■

3.3 A Find the reduced echelon form of A . What is the rank? What is the special solution to $Ax = 0$?

Second differences $-1, 2, -1$
Notice $A_{11} = A_{44} = 1$

$$A = \begin{bmatrix} 1 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 1 \end{bmatrix}$$